

Received 29 November 2023, accepted 22 December 2023, date of publication 1 January 2024,
date of current version 9 January 2024.

Digital Object Identifier 10.1109/ACCESS.2023.3348513

RESEARCH ARTICLE

Multi-Objective Secure Task Offloading Strategy for Blockchain-Enabled IoV-MEC Systems: A Double Deep Q-Network Approach

KOMEIL MOGHADDASI^{ib}, (Member, IEEE), SHAKIBA RAJABI^{ib}, (Member, IEEE),
AND FARHAD SOLEIMANIAN GHAREHCHOPOGH

Department of Computer Engineering, Islamic Azad University, Urmia Branch, Urmia 5756151818, Iran

Corresponding author: Komeil Moghaddasi (k.moghaddasi@ieee.org)

ABSTRACT The Internet of Vehicles (IoV) represents a paradigm shift in vehicular communication, aiming to enhance traffic efficiency, safety, and the driving experience by leveraging interconnected vehicles. Despite its promise, the IoV faces challenges such as efficient task offloading, energy management, and data security. Mobile Edge Computing (MEC) emerges as a solution to some of these challenges by bringing computational resources closer to the vehicular network's edge, yet it raises critical concerns regarding resource management, service continuity, and scalability in dynamic vehicular environments. Addressing both IoV and MEC challenges necessitates robust and dynamic optimization mechanisms. In response to these challenges, our study introduces a multi-objective approach using Double Deep Q-Networks (DDQN). This algorithm combines the strengths of Deep Neural Networks (DNNs) and Deep Learning (DL) techniques, enabling dynamic decision-making that can adapt to changing conditions. By considering multiple objectives, the DDQN algorithm allows for a sophisticated trade-off analysis, efficiently balancing between the different objectives to optimize overall system performance. Through the use of Blockchain technology, known for its secure, decentralized structure, our model enhances the integrity of data, providing a reliable and efficient solution for IoV-MEC systems. We conducted a comparative analysis of our model against the standard Deep Q-Network (DQN) and Deep Deterministic Policy Gradient (DDPG) algorithms, which are prevalent in this field. Our model demonstrated significant improvements over these traditional methods: energy consumption was reduced by 26.4%, latency decreased by 6.87%, and the cost was minimized by 7.41%.

INDEX TERMS Internet of vehicles, mobile edge computing, blockchain, double deep Q-networks, task offloading.

I. INTRODUCTION

The Internet of Things (IoT) has dramatically transformed how we interact with the world around us, ushering in a new era of digital connectivity [1]. Devices ranging from home appliances to industrial machinery are now equipped with sensors, computing capabilities, and network connectivity, enabling them to collect, process, and exchange data [2]. This technological evolution has led to an increased focus on creat-

ing interconnected networks in various sectors, with the IoV emerging as a prominent application of IoT technology [3]. IoV revolutionizes vehicular communications by integrating vehicles with intelligent, networked capabilities, expanding their functionality far beyond traditional transportation [4]. By utilizing cutting-edge technologies such as sensors, cloud computing, and data communication, IoV enables seamless interaction between vehicles, infrastructure, and other smart devices [5]. These developments pave the way for innovative applications such as automated driving, remote vehicle diagnostics, real-time traffic monitoring, enhanced

The associate editor coordinating the review of this manuscript and approving it for publication was Mehdi Sookhak^{ib}.

vehicle safety, and smart transportation systems, among others [6], [7].

These advancements contribute to the creation of smart cities, where digital connectivity and data-driven insights streamline urban living. However, the increasing complexity and expansive capabilities of IoV systems also introduce significant challenges, especially in terms of computational requirements, data offloading, and security [8]. High-speed data processing and decision-making are critical for IoV applications like autonomous driving, which require real-time responses to ensure safety and efficiency [9]. This is where MEC comes into the picture. MEC is a subset of edge computing focused on providing computational resources and services at the edge of mobile networks, catering specifically to mobile devices and applications. Edge computing, in a broader sense, encompasses decentralized processing closer to data sources, serving various devices and applications beyond mobile networks [10], [11]. MEC brings computational resources closer to the edge of the network, essentially near the data source, which drastically reduces latency and improves overall network efficiency [12], [13]. By enabling localized data processing, MEC minimizes network congestion, making it an ideal solution for IoV's extensive data processing needs [14].

However, the implementation of MEC in the IoV domain comes with its own set of challenges. Task offloading, in particular, contends with the unpredictability of network connectivity, the stringent requirements for data privacy, and the complexity involved in harmonizing a range of computational resources [15], [16]. Blockchain technology, renowned for its data security and integrity capabilities, offers a promising solution to these challenges [17], [18]. Incorporating a decentralized network like blockchain in IoT and IoV systems can significantly improve trustworthiness, transparency, and tamper-proof features [19], [20]. The blockchain provides a distributed ledger that records transactions in an immutable manner, ensuring data integrity and protection against unauthorized modifications [21]. This makes blockchain a powerful tool for enhancing the security and efficiency of IoV systems, bringing us closer to realizing the full potential of this transformative technology [22].

Advancing IoV-MEC and blockchain technologies poses a significant challenge in creating an offloading strategy that saves energy, reduces delay, and ensures robust security [23]. Existing methods often compromise one benefit for another, limiting the potential of IoV-MEC systems. Given the large data exchanges in IoV, energy consumption is a critical concern, influencing both sustainability and operational costs. A balanced approach that intelligently minimizes energy use without sacrificing performance and security is vital for the evolution of IoV.

Fig. 1, illustrates the IoV's multi-layered architecture, integrating the physical, network, transport, and application layers, topped by a blockchain layer for secure transactions. Above this, the MEC layer provides localized processing

and storage, optimizing blockchain data handling. The central aim of this paper is to present a unique multi-objective method that optimizes performance while ensuring security. We propose a strategy that uses a DDQN model within a blockchain framework to secure communication channels and improve energy efficiency. This approach aims to balance multiple objectives such as reducing energy consumption, communication overhead, and delays, while also minimizing task offloading costs within an IoV-MEC setting. By considering these multiple objectives, our strategy enables a trade-off analysis, effectively optimizing the overall system performance. The contributions of this study include:

- We present an integrated framework tailored for MEC environments that synergizes cost minimization with the management of computational resources and energy efficiency. This framework is particularly designed to address the unique demands of vehicular networks within the IoV landscape.
- Our study introduces an application of Blockchain technology within the IoV and MEC domains, ensuring enhanced data security and integrity. We detail the customization of Blockchain network architecture to suit the high-mobility environment of IoV, which is a novel contribution to the field.
- We propose a sophisticated offloading strategy that holistically considers and balances latency, energy, and cost efficiency. This is not merely an optimization of existing methods but a comprehensive approach that takes into account the dynamic and heterogeneous nature of IoV networks.
- The study tackles the complex multi-objective optimization problem inherent in IoV systems by leveraging a DDQN-based algorithm. This algorithm is an innovative blend of Q-learning and DNNs, specifically adapted for the real-time decision-making required in IoV offloading scenarios. We demonstrate the novelty of this approach through extensive simulations, showcasing its superior performance in comparison to conventional methodologies.

In Section II, we conduct a thorough examination of pertinent literature, shedding light on prevailing solutions and pinpointing their limitations. Sections III and IV are dedicated to delineating our proposed system model, where we elucidate the intricacies of our data offloading algorithm tailored for IoV environments. Section V is reserved for detailing our experimental methodology, where we describe the setup and metrics used to evaluate our approach. Following this, Section VI presents our empirical findings, offering insights into the efficacy of our proposed model and we engage in a critical discussion, comparing our results with those of existing techniques and highlighting the distinct benefits of our method. Finally, the last Section encapsulates our conclusions, summarizing the key contributions of our work in enhancing Blockchain-Enabled IoV-MEC systems and suggesting avenues for future inquiry.

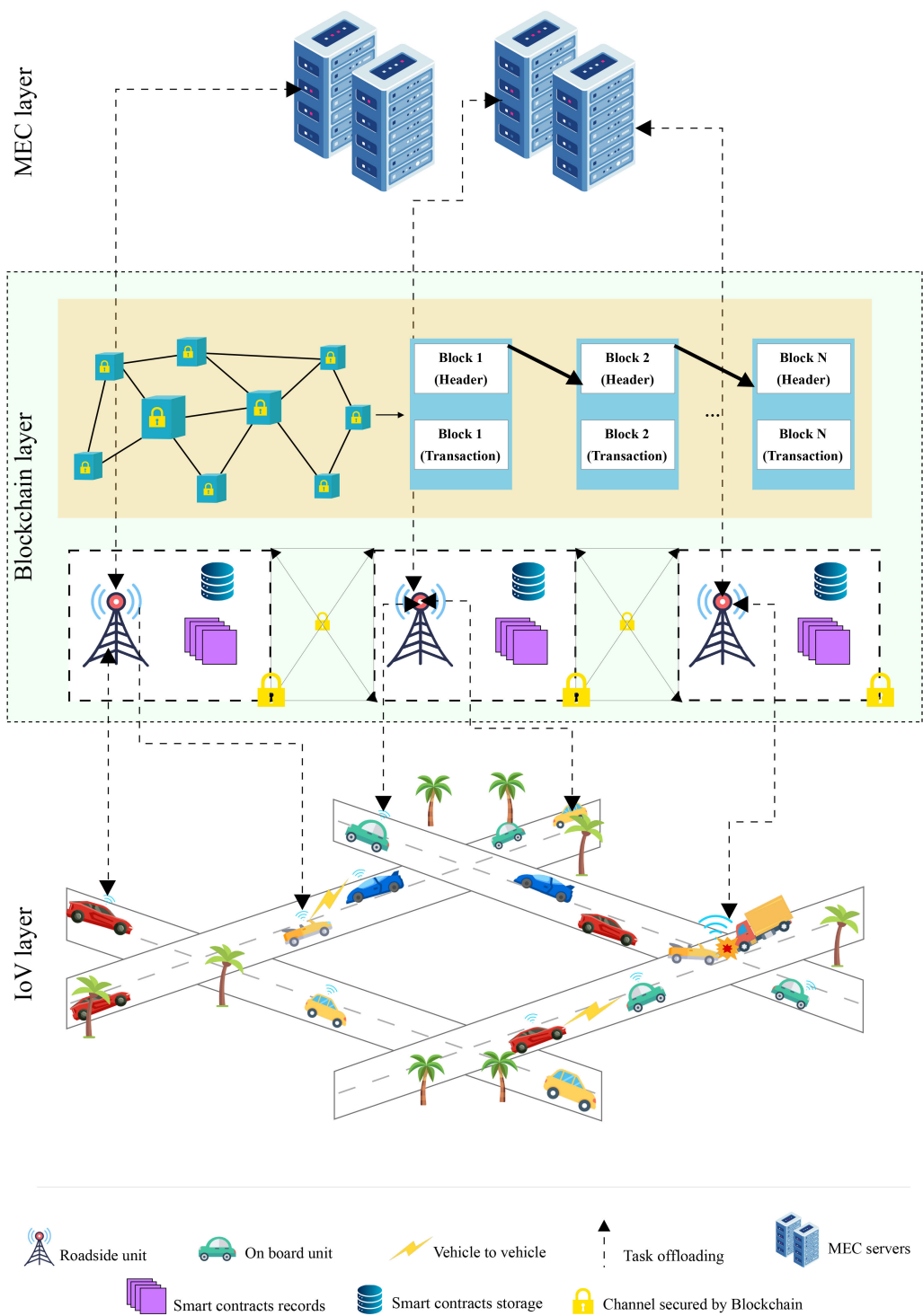


FIGURE 1. Proposed system architecture.

II. RELATED WORKS

In this section, we aim to review related works that revolve around the application of blockchain technology in the IoV paradigm. To start with, In [24], Liu et al. presented a Hierarchical Blockchain-enabled Security threat Assessment

Architecture (HBSAA) for data sharing in IoV smart terminals, using edge and global chains. They employed data virtualization to manage multi-source security threat data and an attribute graph-based metadata model to address complex data relations. An ant colony optimization enhanced

key nodes' query performance. Experiments with large-scale vulnerability data from the National Vulnerability Database validated HBSAA's efficiency in shielding data heterogeneity, enhancing query speed, and reducing service response time.

However, the proposed architecture might incur increased latency for synchronization across the hierarchical blockchain layers, especially in scenarios with high-frequency data updates.

The study by Xiao et al. [25], introduced QcFND, a system utilizing Software-Defined Networking (SDN), edge computing, blockchain, and Bayesian networks. QcFND comprises two layers: edge and vehicles, with the edge including Software-Defined Road Side Units (SDRSUs) housing virtual machines like SDN controllers and blockchain servers. SDN controllers facilitate IoV load balancing, while blockchain servers manage vehicle reports and calculate traffic event probabilities. A Bayesian Network infers the trustworthiness of received reports. Tests on three platforms confirmed QcFND's superior performance compared to other solutions. However, the proposed system may be constrained by the processing capacity of SDRSUs, which could limit the scalability and responsiveness of the QcFND under heavy network traffic conditions.

In [26], Zhang et al. presented a robust framework for data security and privacy in the IoV. They utilized a blockchain-supported access method based on Ciphertext-Policy Attribute-Based Encryption (CP-ABE), with hidden user policies and transactional integrity checks. Local outsourcing technology manages IoV's limited resources during encryption/decryption, and smart contracts minimize human oversight and verification workload. The system supports dynamic attribute updates with forward and backward security, accommodating IoV's dynamic nature. Numerical and security analyses highlighted benefits such as lower storage, time, and computation costs. Nonetheless, Zhang et al.'s approach may be impeded by the computational complexity associated with CP-ABE operations, which could affect system throughput in environments with numerous concurrent transactions.

Li et al. [27], proposed a compact consensus pre-authentication and handover authentication method for IoV, employing blockchain security and decentralization to cut consensus time and computational resources. They tackle high costs from repeated vehicle authentication handovers. The overhead in pre-certification correlates with the number of consensus nodes. Results show reductions of up to 88.14% in vehicle computational overhead, over 60% in RSU overhead, and 71.31% in communication overhead. However, Li et al.'s method may face scalability issues in dense urban environments where the frequency of handovers and the number of consensus nodes can significantly increase.

Lahiri et al. [28] tackle scalability and security challenges in IoV using edge computing and blockchain. The proposed system has regionally independent blockchains recording

the trustworthiness of messages. Vehicle Ad-hoc Networks and edge resources are used for IoV device behavior storage. A hybrid consensus mechanism, combining Proof of Location with Federated Byzantine Agreement, boosts scalability, and security, and reduces block creation time. This energy-optimized approach is eco-friendly and apt for smart city components. Li et al.'s method reduces overhead in IoV authentication, while Lahiri et al.'s system improves scalability and security. Further investigation is needed to understand the impact of consensus nodes on overhead and the practical implementation of these approaches in more complex IoV settings. However, this system may still encounter bottlenecks in inter-regional communication and ledger reconciliation, which could affect its efficiency in larger, more integrated IoV frameworks.

In [29], Zhang et al. proposed the BMEC-FV, a blockchain-driven, multi-access edge computing framework for the future VANET ecosystem. It prioritized communication security in VANETs and optimized throughput and quality of services (QoS) for MEC users. Their approach dealt with the challenge of consensus among blockchain nodes by developing a deep compressed neural network for a Markov decision process, considering various parameters like block size, reliable vehicle features, etc. Simulation results underscored the effectiveness of the proposed ecosystem.

Ye et al. [30] leveraged blockchain technology for data accuracy and reliability in IoV's MEC systems. They introduced a smart resource allocation framework for computation and caching tasks, aiming to optimize the offloading decisions, the number of consensus nodes, block interval, and block size. This approach, modeled as a Markov decision process due to the dynamic resource availability, aimed to reduce energy and computational costs, and increase blockchain throughput. The large-scale optimization problem was addressed using the Asynchronous Advantage Actor-Critic approach. The proposed model outperformed others, yielding a total reward approximately 14% higher than the DQN-based scheme. However, Ye et al.'s framework might face challenges adapting to real-time changes in IoV environments due to the complex nature of the MDP in highly dynamic settings.

Mei et al. [31] propose an efficient and conditional privacy-preserving authentication protocol for blockchain-enabled IoV with edge computing. The protocol combined edge computing and consortium blockchains to support efficient computation and storage, ensuring low communication delay and data auditability. Identity privacy was achieved using a pseudonym mechanism, while vehicle message authentication utilized identity-based signatures. Additionally, clustering selection and key updating algorithms were introduced to handle dynamic vehicle changes and ensure transmission information security. Security analysis and experimental evaluations demonstrated the protocol's effectiveness in terms of communication and computation costs.

In their research [32], Cui et al. proposed CUTE, a containerized edge computing platform, in this study. CUTE facilitated low-latency computation services for IoV by efficiently scheduling containers on edge servers, optimizing computation delay. The platform included a centralized controller for resource management and orchestration and an integrated blockchain for enhanced network security. The researchers formulated vehicle task offloading and container scheduling problems and developed a heuristic algorithm for DAG-based computation tasks. Experiments conducted on the China Mobile Network demonstrated CUTE's ability to provide low-latency computation services for vehicular applications, with the heuristic algorithm outperforming traditional scheduling policies. However, this centralized controller in CUTE could become a single point of failure, which may compromise the system's resilience and scalability in extensive IoV scenarios.

In their study, Liao et al. [33], presented an intelligent task offloading framework for the IoV, enhancing fairness and security via blockchain and smart contracts. A trust metric and assessment mechanism inform the QUOTA-UCB algorithm, optimizing offloading and handover decisions against delays and costs. Validation through analysis and simulations confirms the framework's efficacy and reliability in IoV applications, establishing its potential for improving the network's operational dynamics. However, the proposed framework may face increased overhead for trust metric computation and smart contract execution, which could affect the system's efficiency during peak IoV traffic periods.

In [34], Iqbal et al. introduced a secure fog computing framework utilizing Roadside Units (RSUs) to delegate tasks to proximate fog vehicles, guided by reputation scores recorded on a distributed blockchain ledger. Empirical outcomes exhibit notable enhancements in queuing time, end-to-end delay, and task fulfillment efficiency, surpassing the standard queuing-based offloading methods. Despite this, the presented framework might confront scalability issues with the blockchain ledger as the number of transactions increases, potentially leading to longer verification times and delays in task delegation.

Zheng et al. in [35], investigated vehicular safety and data offloading in a cloud blockchain-enabled ECCO system. They proposed a secure SDVs framework to achieve vehicular consensus and a blockchain-based access control to safeguard against unauthorized offloading. Optimizing offloading decisions, consensus mechanisms, resource allocation, and bandwidth, their method minimizes system delays, energy, and costs. A Deep Reinforcement Learning (DRL) algorithm using extended deep Q-networks was developed to enhance offloading efficiency. Numerical simulations showcased their framework's superiority in access control and offloading, offering significant benefits over traditional approaches.

In academic circles, the combination of the IoV with MEC and blockchain technology is recognized for its potential to

enhance data management, increase processing speeds, and strengthen security within intelligent transportation systems. However, the development of an effective task offloading strategy that simultaneously reduces energy consumption, decreases latency, and lowers costs still presents a significant challenge. Previous research contributions have advanced individual components of this integrated system, yet a unified solution addressing all three objectives collectively remains absent.

Our research introduces an approach that centers on multi-objective optimization for IoV applications supported by MEC within a blockchain network. Instead of traditional algorithms that focus on a single aspect, our method is designed to optimize a range of crucial objectives: conserving energy, minimizing latency, and reducing costs. This approach aims to achieve a balanced and efficient operation of the IoV ecosystem, ensuring that the system's performance is optimized in terms of energy, time, and financial resources.

This comprehensive strategy is poised to significantly advance the operational effectiveness of IoV applications, marking a valuable addition to the academic literature in this field. Table 1 summarizes the characteristics of related works and this paper.

III. SYSTEM MODEL AND OFFLOADING STRATEGY

In this section, we aim to provide a comprehensive mathematical exposition of our system model, meticulously detailing the architecture of the IoV, MEC, and blockchain network.

A. IoV COMMUNICATIONS

Fig. 2, illustrates the diverse communication models within the IoV ecosystem, such as Vehicle-to-Grid (V2G) [36], Vehicle-to-Pedestrian (V2P) [37], Vehicle-to-Everything (V2X) [38], Vehicle-to-Vehicle (V2V) [39], and Vehicle-to-Infrastructure (V2I) [40], each facilitating distinct and critical interactions within the network. To accurately model these interactions, our system incorporates specific protocols and technologies tailored to each communication type. V2G interactions are modeled to reflect energy exchange scenarios and their impact on the grid and vehicle operations. V2P communications are considered for safety applications, integrating pedestrian movements and alert systems. V2X is modeled as an overarching framework that includes and extends beyond V2G and V2P, to cover comprehensive vehicular communication scenarios.

V2V and V2I models are designed to capture the dynamic information exchange between vehicles and the road infrastructure, respectively. These models take into account the mobility patterns, network conditions, and protocol-specific characteristics to ensure a realistic and interconnected IoV simulation environment.

B. SYSTEM MODEL

In this section, we introduce the fundamental assumptions that underpin our proposed method for task offloading in the IoV. Our system model is predicated on a stable IoV

TABLE 1. Properties of the related schemes and the proposed scheme.

Ref	Main Contribution	Application and Environment						Approach	
		Blockchain	IoV	MEC	Fog computing	Edge computing	Cloud computing	End device	End server
[24]	Hierarchical Blockchain-enabled Security Threat Assessment Architecture for IoV, optimizing data sharing and query efficiency.	✓	✓			✓			✓
[25]	Introduced QcFND, enhancing IoV with SDN, edge computing, blockchain, and Bayesian networks, ensuring superior performance.	✓	✓			✓			✓
[26]	Proposed a robust IoV data security framework using blockchain-supported CP-ABE, balancing efficiency and computational complexities.	✓	✓				✓		✓
[27]	Proposed a streamlined IoV consensus and handover authentication method using blockchain, reducing computational overhead significantly.	✓	✓						
[28]	Addressed IoV scalability and security using edge computing and blockchain, employing regional blockchains and a hybrid consensus mechanism for efficiency.	✓	✓			✓			✓
[29]	Proposed BMEC-FV, a blockchain-driven framework optimizing VANET security, throughput, and MEC QoS.	✓	✓	✓					✓
[30]	Utilized blockchain for IoV MEC data accuracy, introducing a smart allocation framework optimizing decisions, consensus nodes, and energy costs.	✓	✓	✓					✓
[31]	Proposed a privacy-preserving authentication protocol for blockchain-enabled IoV with edge computing, ensuring efficiency and security in communication and computation.	✓	✓			✓			✓
[32]	Introduced CUTE, a containerized edge computing platform, optimizing IoV computation delay via container scheduling, with blockchain for enhanced security.	✓	✓			✓			✓
[33]	Introduced an intelligent task offloading framework for IoV, enhancing fairness and security through blockchain and smart contracts.	✓			✓				
[34]	Introduced a secure fog computing framework using RSUs, delegating tasks based on blockchain-guided reputation scores and improving efficiency.	✓			✓			✓	
[35]	Introduced a secure SDVs framework for vehicular safety and data offloading, showcasing superiority over traditional approaches.	✓				✓	✓		✓
This Paper	Introduced a multi-objective approach for dynamic decision-making in IoV-MEC systems. Enhanced by Blockchain, achieving significant improvements over traditional methods.	✓	✓	✓					✓

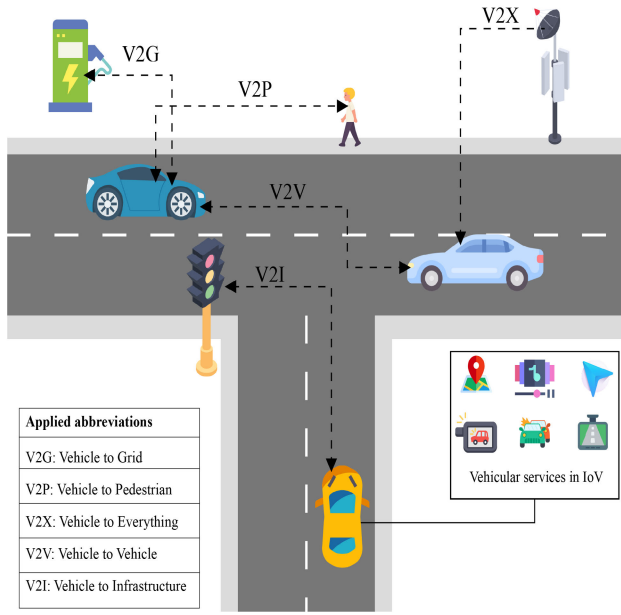


FIGURE 2. Vehicular communications.

environment where vehicles are equipped with robust communication technologies, ensuring reliable connectivity with RSUs for seamless data transmission.

We posit that the network topology remains quasi-static, characterized by predictable vehicular mobility patterns that facilitate efficient routing protocols and minimal packet loss. The tasks designated for offloading are assumed to be latency-sensitive, necessitating adherence to strict, yet realistic deadlines. These tasks are quantified in terms of computational cycles, which represent the processing demands imposed on the system. In parallel, the computing capacity of both in-vehicle devices and edge nodes is denoted in cycles per second (Hertz), delineating the maximum processing power available for task execution at any moment. We further postulate that the computational resources, albeit limited, are adequately provisioned to accommodate the simultaneous processing demands of multiple tasks without substantial queuing delays. The allocation of these resources is reflective of a typical urban IoV scenario, with a balanced distribution between the vehicles and RSUs. Channel conditions within this model are considered to be well-known and consistent throughout task transmissions, enabling precise estimations of communication delays. Central to our assumptions is the expectation that the DDQN algorithm utilized for decision-making has access to a historical dataset, which informs the training process. This model intricately connects vehicles, RSUs, and edge nodes, ensuring synchronized task offloading and optimal resource utilization via a trained DDQN algorithm.

To model the task offloading process in an IoV architecture, we take into account different elements: vehicles (V), Edge Nodes (EN), tasks (T), and a set of parameters that control the process.

Let τ_{vt} be a binary variable representing the offloading decision according to (1).

$$\tau_{vt} \in \{0, 1\}, \forall v \in V, \forall t \in T \quad (1)$$

where $\tau_{vt}=1$ indicates task t is offloaded from vehicle v , otherwise, it is processed locally.

The energy consumption for task execution ϵ_{vt} by vehicle v performing task t , adopts a cubic relationship between CPU frequency and energy, reflecting the dynamics of vehicular computational energy, as given by (2).

$$\epsilon_{vt} = \kappa_v \times f_{vt}^2 \times \tau_{vt} \quad (2)$$

where κ_v denotes the energy coefficient of vehicle v and f_{vt} is the CPU cycle frequency required for task t .

The aggregate energy consumption Ξ for the IoV system is then referred to (3).

$$\Xi = \sum_{v \in V} \sum_{t \in T} (\epsilon_{vt}) \quad (3)$$

Here, V represents the set of all vehicles, and T denotes the set of all tasks within the IoV system. This equation integrates the individual energy expenditures into a total value that reflects the energy requirements of the entire system.

The Data Offloading Ratio (DOR) φ represents the fraction of tasks offloaded from vehicles to external processing units. It is defined as the ratio of offloaded tasks to the total potential offloading scenarios, calculated as (4).

$$\varphi = \frac{\sum_{v \in V} \sum_{t \in T} (\tau_{vt})}{(|V| \times |T|)} \quad (4)$$

Here, τ_{vt} is a binary indicator of whether task t from vehicle v is offloaded (1) or not (0), $|V|$ is the total number of vehicles, and $|T|$ denotes the total number of tasks.

The energy efficiency ϑ is modeled as (5):

$$\vartheta = \sum_{v \in V} \sum_{t \in T} \left(\frac{c_{vt}}{\epsilon_{vt}} \right) \quad (5)$$

where c_{vt} denotes the computation capacity for task t in vehicle v . We should mention that peak capacity is a vehicle's maximum constant computational power, while instantaneous capacity varies with the current load and ongoing processes.

The computation latency λ_{vt} of task t in vehicle v , which measures the time taken for vehicles to compute tasks is shown in (6).

$$\lambda_{vt} = \delta_{vt} \times \tau_{vt} \quad (6)$$

where δ_{vt} is the computation time required for task t in vehicle v . The total latency Λ for the system is referred to (7).

$$\Lambda = \sum_{v \in V} \sum_{t \in T} (\lambda_{vt}) \quad (7)$$

The total cost ζ , considering both energy and latency, encapsulates the trade-off between energy efficiency and processing speed. It is calculated as (8).

$$\zeta = \alpha \times \Xi + \beta \times \Lambda \quad (8)$$

In this formula, α and β are tunable weighting factors that prioritize energy and latency, respectively. Ξ denotes the overall energy consumed by the system, and Λ signifies the cumulative latency incurred across all tasks. Our objective function minimizes this total cost, reflecting an optimization balance between energy savings and timely data processing.

$$\min \zeta \quad (9)$$

subject to computation capacities and energy constraints. This is formulated as an optimization problem according to (10).

$$\min \sum_{v \in V} \sum_{t \in T} [\alpha \times \epsilon_{vt} + \beta \times \lambda_{vt}] \quad (10)$$

subject to $c_{vt} \leq C_{max}$, $\forall v \in V, \forall t \in T$, and $\epsilon_{vt} \leq E_{max}$, $\forall v \in V, \forall t \in T$.

C. MEC ENVIRONMENT

For the MEC architecture, we consider the same vehicles V , tasks T and edge nodes EN . Let f_{en} be the CPU frequency of the edge node en , c_{ent} be the computation capacity of edge node en for task t , and ϵ_{ent} be the energy consumption of edge node en for task t .

According to (11), we denote by σ_{ent} a binary variable that signifies whether task t from any vehicle is offloaded to edge node en :

$$\sigma_{ent} \in \{0, 1\}, \forall en \in EN, \forall t \in T \quad (11)$$

The computation time for task t on edge node en , δ_{ent} , which represents the duration to process tasks is given by (12).

$$\delta_{ent} = \sqrt{\left(\frac{1}{f_{en}}\right) \times \tau_{vt}}, \forall v \in V, \forall t \in T, \forall en \in EN \quad (12)$$

where τ_{vt} is the size of task t from vehicle v .

Energy consumption at edge node en for task t , ϵ_{ent} , which denotes energy used by edge node is modeled as (13).

$$\epsilon_{ent} = \int_0^{\tau_{vt}} k_{en} \times f_{en}^2 dt, \forall v \in V, \forall t \in T, \forall en \in EN \quad (13)$$

where k_{en} is the energy coefficient for edge node en .

The aggregate energy consumption Ξ_{en} for all edge nodes, which signifies the total energy used by all edge nodes combined is calculated as (14).

$$\Xi_{en} = \sum_{en \in EN} \sum_{t \in T} (\epsilon_{ent}) \quad (14)$$

The task processing latency λ_{ent} at edge node en for task t is expressed as (15).

$$\lambda_{ent} = \delta_{ent} \times \sigma_{ent}, \forall en \in EN, \forall t \in T \quad (15)$$

The total latency Λ_{en} across all edge nodes is defined as (16).

$$\Lambda_{en} = \sum_{en \in EN} \sum_{t \in T} (\lambda_{ent}) \quad (16)$$

Energy efficiency ϑ_{en} at an edge node en is defined as (17).

$$\vartheta_{en} = \prod_{t \in T} \left(\frac{c_{ent}}{\epsilon_{ent}} \right), \forall en \in EN \quad (17)$$

The total cost ζ_{en} , considering both energy and latency, at edge node en is expressed as (18).

$$\zeta_{en} = \alpha \times \Xi_{en} + \beta \times \Lambda_{en} \quad (18)$$

where α and β are the weighting parameters for energy and latency respectively.

Our objective is to minimize the total cost as calculated in (19).

$$\min \sum_{en \in EN} (\zeta_{en}) \quad (19)$$

subject to computation capacities and energy constraints. This can be formulated as an optimization problem as shown in (20).

$$\min \sum_{en \in EN} \sum_{t \in T} [\alpha \times \epsilon_{ent} + \beta \times \lambda_{ent}] \quad (20)$$

subject to $c_{ent} \leq C_{max}$, $\forall v \in V, \forall t \in T$, and $\epsilon_{ent} \leq E_{max}$, $\forall v \in V, \forall t \in T$

D. BLOCKCHAIN NETWORK ARCHITECTURE

In the proposed blockchain framework for the IoV, we utilize a private blockchain structure as depicted in Fig. 3, and a consensus mechanism with select consensus nodes [41]. These nodes are carefully chosen based on their stability and processing ability to ensure transaction integrity and maintain the blockchain's reliability. Our design incorporates Identity Authentication by blending Anonymous Authentication, Public Key Infrastructure (PKI), and crowd-sourcing, striking a balance between user privacy and system security. The PKI acts as a protective shield for digital identities, while Crowdsourcing utilizes the collective power of the network to enhance security measures. To safeguard privacy, we employ Location-based, Content-based, and Identity-based methods. These protect users' locations, secure the confidentiality of data during offloading tasks, and conceal identities, enhancing user privacy within the network. Also, in this framework, the data stored primarily includes transaction records, authentication logs, and task offloading events. For Access Control, we have established Trust-based, Data-based, and Risk-based protocols.

Trust-based protocols encourage a cooperative network environment, Data-based protocols protect sensitive data, and Risk-based protocols scrutinize each access request to minimize security threats. Our comprehensive approach to blockchain in IoV environments emphasizes not just security and privacy, but also efficient and scalable consensus operations. This ensures that our system can meet the unique demands of the IoV, offering a secure and dependable network framework.

1) ENERGY AND COST-EFFICIENT BLOCKCHAIN INTEGRATION FRAMEWORK

For modeling the blockchain network architecture, we incorporate edge nodes EN , a set of blocks B , and transactions TR . Let's represent h_b as the block height, d_{tr} as the data size of

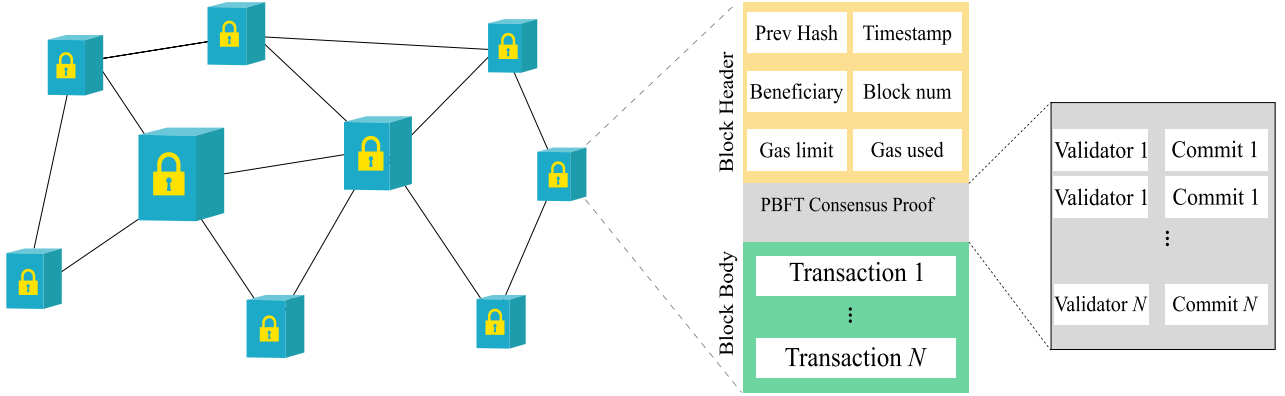


FIGURE 3. Private blockchain structure.

transaction tr , p_b as the propagation delay of block b , and t_b as the block generation time.

The block generation time t_b for block, b expressed as (21).

$$t_b = \sum_{tr} \in TR \left(\frac{d_{tr}}{f_{en}} \right) + \sqrt{h_b}, \forall b \in B, \forall en \in EN \quad (21)$$

The propagation delay p_b for block b across the network, which measures time taken for blocks to travel network-wide is expressed as (22).

$$p_b = \int_0^{t_b} \gamma \times dt, \forall b \in B \quad (22)$$

where γ is a propagation constant.

The energy consumed ϵ_b to generate block b on edge node en is defined as (23).

$$\epsilon_b = \sum_{tr} \in TR \kappa_{en} \times \left(\frac{d_{tr}}{f_{en}} \right)^2, \forall b \in B, \forall en \in EN \quad (23)$$

where κ_{en} is the energy coefficient for edge node en .

According to (24), the total energy consumption Ξ_b for the blockchain network is:

$$\Xi_b = \sum_{vb} \in B (\epsilon_b) \quad (24)$$

Transaction validation time v_{tr} for transaction tr on edge node en , which denotes the duration to validate transactions on nodes is referred to (25).

$$v_{tr} = \sqrt{\left(\frac{d_{tr}}{f_{en}} \right)}, \forall tr \in TR, \forall en \in EN \quad (25)$$

The total transaction validation time V across the blockchain network is defined in (26).

$$V = \sum_{tr} \in TR (v_{tr}) \quad (26)$$

The total cost ζ_b considering energy consumption, block generation time, and propagation delay is shown in (27).

$$\zeta_b = \alpha \times \Xi_b + \beta \times t_b + \gamma \times p_b, \forall b \in B \quad (27)$$

where α , β , and γ are the weighting parameters.

We aim to minimize the total cost refer to (28):

$$\min \sum_{vb} \in B (\zeta_b) \quad (28)$$

subject to energy constraints and network capacity.

This formulated as an optimization problem as expressed (29).

$$\min \sum_{vb} \in B [\alpha \times \epsilon_b + \beta \times t_b + \gamma \times p_b] \quad (29)$$

subject to $\epsilon_b \leq E_{max}, \forall b \in B$.

E. OFFLOADING STRATEGY

The offloading strategy involves the decisions made about whether to offload tasks from vehicles to edge nodes or process them locally, based on different factors such as the load on the edge nodes, the size of tasks, and the available resources. We have already introduced the sets of vehicles V , tasks T , and edge nodes EN . Let's further denote l_{en} as the load on edge node en , s_{vt} as the size of task t from vehicle v , and r_{en} as the available resources on edge node en .

An offloading decision ω_{vten} represented as a binary variable refer to (30).

$$\omega_{vten} \in \{0, 1\}, \forall v \in V, \forall t \in T, \forall en \in EN \quad (30)$$

The load l_{en} on an edge node en , which represents the cumulative task demands on edge nodes is given by (31).

$$l_{en} = \sum_{vv} \in V \sum_{vt} \in T (\omega_{vten} \times s_{vt}), \forall en \in EN \quad (31)$$

The utilization u_{en} of edge node en calculated as (32).

$$u_{en} = \frac{l_{en}}{r_{en}} \forall en \in EN \quad (32)$$

The total energy consumption Ξ_{vten} for the offloading decision ω_{vten} is expressed as (33).

$$\Xi_{vten} = \int_0^{l_{en}} \kappa_{en} \times dt, \forall v \in V, \forall t \in T, \forall en \in EN \quad (33)$$

Equation (34) can calculate the latency λ_{vten} for offloading decision ω_{vten} .

$$\lambda_{vten} = \sqrt{\left(\frac{l_{en}}{f_{en}} \right)}, \forall v \in V, \forall t \in T, \forall en \in EN \quad (34)$$

The total latency Λ_{vten} for all offloading decisions is expressed as (35).

$$\Lambda_{vten} = \sum_{v \in V} \sum_{t \in T} \sum_{en \in EN} (\lambda_{vten}) \quad (35)$$

The offloading strategy's effectiveness ψ_{vten} calculated as (36).

$$\psi_{vten} = \prod_{v \in V} \prod_{t \in T} \prod_{en \in EN} \left(\frac{u_{en}}{\Xi_{vten}} \right) \quad (36)$$

The total cost ζ_{vten} for the offloading decision ω_{vten} , considering energy and latency, is referred to (37).

$$\zeta_{vten} = \alpha \times \Xi_{vten} + \beta \times \lambda_{vten}, \forall v \in V, \forall t \in T, \forall en \in EN \quad (37)$$

where α and β are the weighting parameters for energy and latency respectively.

Our objective is to maximize effectiveness while minimizing the total cost as calculated in (38).

$$\max \psi_{vten}, \min \zeta_{vten} \quad (38)$$

subject to computation capacities and energy constraints.

This is formulated as an optimization problem that is calculated in (39).

$$\begin{aligned} & \max \prod_{v \in V} \prod_{t \in T} \prod_{en \in EN} \left(\frac{u_{en}}{\Xi_{vten}} \right), \\ & \min \sum_{v \in V} \sum_{t \in T} \sum_{en \in EN} [\alpha \times \Xi_{vten} + \beta \times \lambda_{vten}] \end{aligned} \quad (39)$$

subject to $l_{en} \leq r_{en}, \forall en \in EN$, and $\Xi_{vten} \leq E_{max}, \forall v \in V, \forall t \in T, \forall en \in EN$.

IV. RESEARCH PROBLEM FORMULATION AND PROPOSED ALGORITHM

In this section, we outline the computational challenges in IoV and propose an algorithm for efficient task offloading and resource management.

A. PROBLEM FORMULATION

Having established the offloading strategy, we now model the problem mathematically by framing it as an optimization challenge.

We aim to balance task offloading with factors like energy consumption, latency, and system efficiency in the IoV, MEC, and blockchain contexts. Let ϕ represent the overall system performance metric that we aim to optimize, which is a function of energy efficiency, data offloading ratio, block generation time, and transaction validation time.

The overall system performance ϕ is defined as (40).

$$\phi = \alpha \times \eta_{eff} + \beta \times \rho_{off} - \gamma \times \theta_{gen} - \delta \times \nu_{val} \quad (40)$$

where η_{eff} is energy efficiency, ρ_{off} is data offloading ratio, θ_{gen} is block generation time, ν_{val} is transaction validation time, and α, β, γ , and δ are corresponding weights.

The energy efficiency η_{eff} is defined as (41).

$$\eta_{eff} = \frac{\sum_{v \in V} \sum_{t \in T} \sum_{en \in EN} (\omega_{vten} \times \Xi_{vten})}{\sum_{en \in EN} (\epsilon_{en} + \Xi_b)} \quad (41)$$

The DOR ρ_{off} is defined as (42).

$$\rho_{off} = \frac{\sum_{v \in V} \sum_{t \in T} \sum_{en \in EN} (\omega_{vten} \times s_{vt})}{\sum_{v \in V} \sum_{t \in T} (s_{vt})} \quad (42)$$

The average block generation time θ_{gen} for all blocks is shown in (43).

$$\theta_{gen} = \frac{\sum_{b \in B} (t_b)}{|B|} \quad (43)$$

The average transaction validation time ν_{val} for all transactions is achieved by expressing it through (44).

$$\nu_{val} = \frac{\sum_{tr \in TR} (v_{tr})}{|TR|} \quad (44)$$

The objective is to maximize ϕ refer to (45).

$$\max \phi \quad (45)$$

subject to offloading decision constraints, energy consumption limits, and network capacities.

This can be formulated as an optimization problem as presented in (46).

$$\max [\alpha \times \eta_{eff} + \beta \times \rho_{off} - \gamma \times \theta_{gen} - \delta \times \nu_{val}] \quad (46)$$

subject to (47).

$$\begin{aligned} & l_{en} \leq r_{en}, \text{ for all } en \in EN, \epsilon_{en} + \Xi_b \leq E_{max}, \forall en \in EN, \\ & b \in B, \text{ and } \sum_{v \in V} \sum_{t \in T} \sum_{en \in EN} (\omega_{vten} \times s_{vt}) \leq \sum_{v \in V} \sum_{t \in T} (s_{vt}) \end{aligned} \quad (47)$$

The constraints can be further modeled as (48), for the load of each edge node.

$$\sum_{v \in V} \sum_{t \in T} (\omega_{vten} \times s_{vt}) \leq r_{en}, \forall en \in EN \quad (48)$$

For energy consumption refer to (49).

$$\epsilon_{en} + \Xi_b \leq E_{max}, \forall en \in EN, b \in B \quad (49)$$

For data offloading according to (50).

$$\begin{aligned} & \sum_{v \in V} \sum_{t \in T} \sum_{en \in EN} (\omega_{vten} \times s_{vt}) \\ & \leq \sum_{v \in V} \sum_{t \in T} (s_{vt}) \end{aligned} \quad (50)$$

To accommodate the various parameters, the energy functions consumed by each edge node en is given by (51).

$$\epsilon_{en} = \sum_{v \in V} \sum_{t \in T} (\omega_{vten} \times \kappa_{vt} \times s_{vt}) + \zeta \times t_{on} \quad (51)$$

where κ_{vt} is the energy consumption for task t on vehicle v , ζ is the idle power of the edge node, and t_{on} is the time it is on.

The energy consumed for block generation is given by (52).

$$\Xi_b = \sum_{\forall tr} \in TR(\xi_{tr}) \quad (52)$$

where ξ_{tr} is the energy consumption for transaction tr .

Finally, the optimization problem can be represented as (53).

$$\max \phi = \alpha \times \eta_{eff} + \beta \times \rho_{off} - \gamma \times \theta_{gen} - \delta \times v_{val} \quad (53)$$

B. PROPOSED DDQN ALGORITHM

There has been growing interest in applying DDQN algorithms for task offloading optimization and related fields, such as [42], [43], [44], and [45]. To solve the multi-objective optimization problem identified in the IoV architecture, MEC environment, and the blockchain network, we introduce a DDQN-based [46] algorithm. DDQN, an extension of DQN, is an algorithm that combines Q-learning with DNN. It leverages the strength of DL with DNNs to approximate the Q-function and applies a unique methodology to estimate the maximum Q-value for the next state, thereby alleviating the over-estimation issue common in traditional Q-learning.

DDQN algorithm as shown in Fig. 4, is composed of two identical DNNs: the primary network and the target network. The primary network learns and updates its parameters after each iteration, while the target network updates its parameters less frequently, typically after a fixed number of iterations. The target network provides the Q-value for the next stage in our Q-function approximation, while the primary network is used to select the action that maximizes that value. By decoupling the action selection from the Q-value generation, we can reduce the over-optimistic Q-value estimates that result from the maximization step. The DDQN algorithm begins by observing the current state of the environment, which includes parameters like the tasks to be offloaded from vehicles, the status and load of the MEC edge nodes, the block generation time, the transaction validation time, and the current energy consumption. Each of these parameters constitutes an aspect of the state representation in our DL model. Actions in our model correspond to the decisions about whether to offload a particular task to a MEC node or process it within the vehicle. The DDQN algorithm selects actions based on an ϵ -greedy policy, which means it chooses the action that it currently estimates to have the highest Q-value with probability $1 - \epsilon$, and a random action with probability ϵ . This strategy allows the algorithm to explore the action space and avoid getting stuck in sub-optimal solutions. After taking an action, the algorithm receives a reward and observes the new state of the system.

The reward is a measure of how beneficial the action was, computed as a combination of factors including reduction in energy consumption, improved system efficiency, minimized latency, and successful task completion. The algorithm uses this information - the current state, the action taken, the reward received, and the new state observed - to update the

Q-value of the action taken in the current state. The updating process involves calculating the difference between the estimated Q-value and the observed reward plus the estimated maximum future Q-value, then adjusting the parameters of the primary network to reduce this difference.

Algorithm 32, demonstrates a comprehensive explanation of how the proposed algorithm is trained and works. Our DDQN algorithm learns over time by iterating through this process, gradually improving its understanding of the environment and the impact of its actions. It leverages the power of neural networks to manage the complexity of the IoV, MEC, and blockchain systems, enabling efficient task offloading while minimizing energy consumption and latency and maximizing system efficiency.

C. COMPLEXITY ANALYSIS

To ensure the viability of our DDQN-based task offloading strategy in a practical scenario, it is essential to understand its computational complexity. In this section, we analyze the time complexity and space complexity of our proposed algorithm. The time complexity of our DDQN algorithm is primarily determined by three components: the forward propagation to compute the Q-value, the backpropagation to update the weights of the neural network, and the experience replay procedure.

The time complexity of the forward propagation is given by (54).

$$T_{fp} = O\left(\sum_i = 1^n (d_i \times d_{i-1})\right) \quad (54)$$

where d_i represents the number of neurons in the i^{th} layer, and n is the number of layers in the neural network.

The time complexity of the backpropagation is approximately the same as the forward propagation as calculated in (55).

$$T_{bp} = O\left(\sum_i = 1^n (d_i \times d_{i-1})\right) \quad (55)$$

The time complexity of the experience replay procedure, which involves updating the Q-values based on a mini-batch of transitions, is given by (56).

$$T_{er} = O\left(m \times \sum_i = 1^n (d_i \times d_{i-1})\right) \quad (56)$$

where m is the size of the mini-batch.

Hence, the total time complexity of the DDQN algorithm in one iteration is expressed in (57).

$$\begin{aligned} T_{total} &= O(T_{fp} + T_{bp} + T_{er}) \\ &= O\left(2 \times \sum_i = 1^n (d_i \times d_{i-1}) + m \right. \\ &\quad \left. \times \sum_i = 1^n (d_i \times d_{i-1})\right) \end{aligned} \quad (57)$$

The space complexity of the DDQN algorithm is determined by the memory required to store the weights and biases of

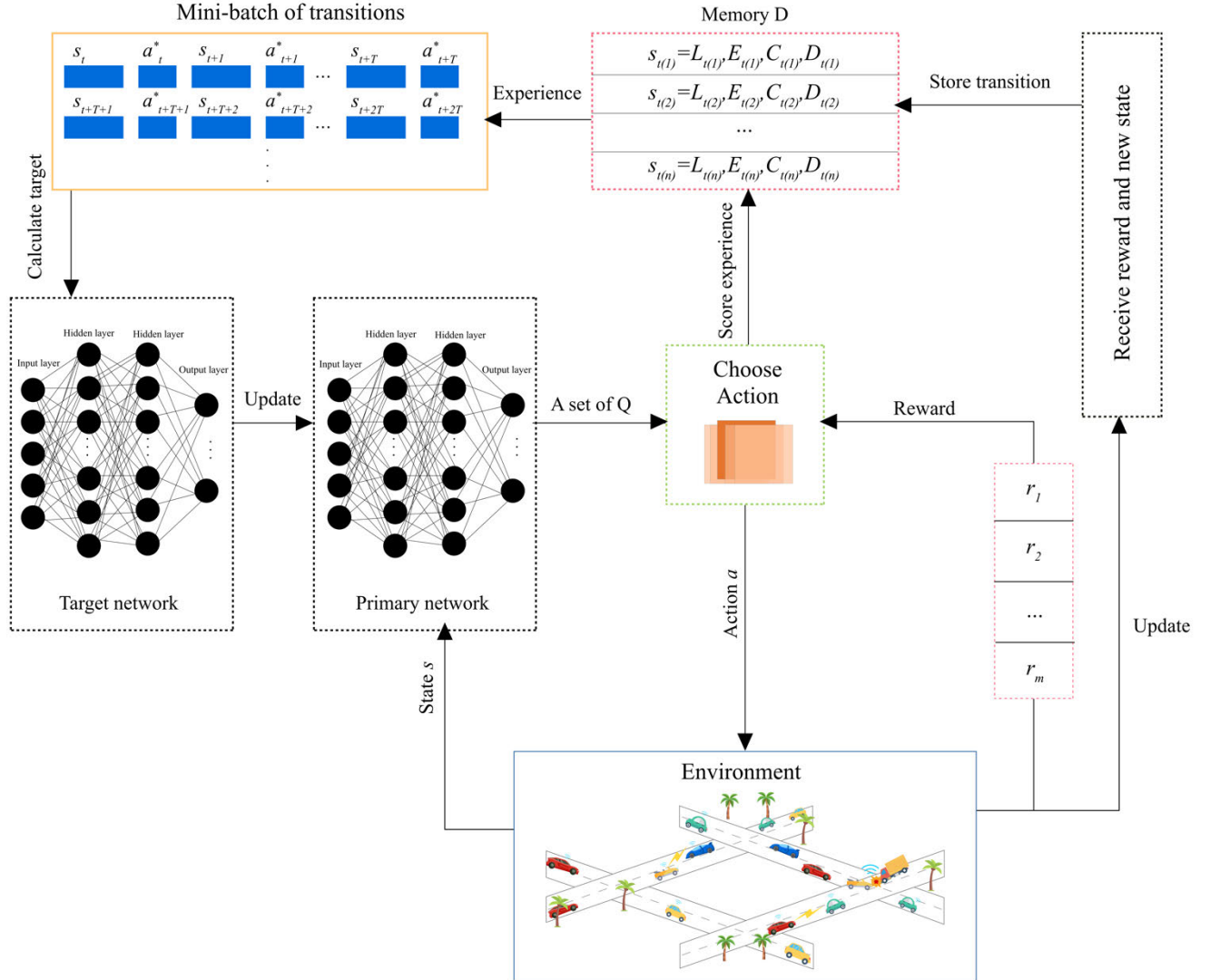


FIGURE 4. Proposed DDQN algorithm architecture.

the neural network, and the memory required to store the transitions in the replay buffer.

The space complexity due to the weights and biases is calculated in (58):

$$S_{wb} = O\left(\sum_i = 1^n (d_i \times d_{i-1} + d_i)\right) \quad (58)$$

The space complexity due to the replay buffer is accomplished refer to (59).

$$S_{rb} = O(b \times (s_s + s_a + s_r + s'_s)) \quad (59)$$

where b is the capacity of the replay buffer, and s_s , s_a , s_r , and s'_s are the sizes of the state, action, reward, and next state, respectively. Thus, the total space complexity of the DDQN algorithm is expressed in (60).

$$\begin{aligned} S_{total} &= O(S_{wb} + S_{rb}) \\ &= O\left(\sum_i = 1^n (d_i \times d_{i-1} + d_i)\right) \end{aligned}$$

$$+ b \times (s_s + s_a + s_r + s'_s)) \quad (60)$$

Given this, the time complexity and space complexity of our DDQN-based task offloading strategy is both linear concerning the size of the neural network and the capacity of the replay buffer, making it efficient and scalable for practical applications in the IoV, MEC, and blockchain context.

V. EXPERIMENTAL SETUP

To validate the performance and efficacy of our model, a comprehensive series of experiments were designed and implemented. These experiments simulated a vehicular network environment to test our proposed DDQN model and compare its performance with standard DQN, DDPG, and random offloading models. The tests were conducted on a robust computing system equipped with an Intel i9-7920X CPU at 4.3 GHz, 64 GB DDR4 2400MT/s RAM, and dual Nvidia RTX 2080 Ti 11 GB GPUs, guaranteeing consistent

Algorithm 1 Offloading strategy using DDQN

```

1 Initialize:
2 State space S
3 Action space A
4 Primary network  $Q_{\text{primary}}$  with random weights w
5 Target network  $Q_{\text{target}}$  with weights, w-set equal
  to the
6 primary network
7 Memory D to store transitions
8 Set exploration rate  $\varepsilon = 1.0$ 
9 Set discount factor  $\gamma = 0.9$ 
  Set the number of episodes N
10 for episode = 1 to N do:
11   Initialize the environment, get the initial
    state s
12   Set the total reward for this episode to zero
13   for each time step in the episode do:
14     Choose action a:
15     if random number  $< \varepsilon$  then:
16       a = random choice from
17     else:
18       a =  $\text{argmax}_a Q_{\text{primary}}(s, a)$ 
19     Execute action a in the environment
20     Receive reward r and new state s'
21     Store transition (s, a, r, s') in memory D
22     Sample random mini-batch of transitions
      ( $s_j, a_j, r_j, s'_j$ ) from D
23     Calculate target  $y_j$  for each sampled
      transition:
24     |  $y_j = r_j + \gamma * \max_{a'} Q_{\text{target}}(s'_j, a')$ 
25     Update  $Q_{\text{primary}}$  using ( $s_j, a_j$ ) and  $y_j$ :
26     | Perform a gradient descent step on
      |  $(y_j - Q_{\text{primary}}(s_j, a_j))^2$  concerning
      | the network parameters
27     Every C steps, update  $Q_{\text{target}} = Q_{\text{primary}}$ 
28     Update the state:  $s = s'$ 
29     Decrease  $\varepsilon$ ,  $\varepsilon = \varepsilon - 0.001$ , until it reaches
      a minimum value
30   if episode % some fixed number == 0:
31     Report average reward per episode so far
32 return  $Q_{\text{primary}}$ 

```

and powerful computational capacity for the training and evaluation of the models.

A. EVALUATION METRICS

In our experiments, we assessed the models using the following metrics: Energy consumption, Energy efficiency, Data Offloading Ratio, Latency, Cost, Block Generation Time, Transaction Validation Time, and Reward. These metrics offer a comprehensive view of the system's performance, providing critical insights into energy efficiency, system latency,

cost-effectiveness, data integrity, and the overall reward generated by the model.

B. DATASET AND SIMULATION ENVIRONMENT

In this study, we investigate the integration of the IoV within a MEC system, underpinned by blockchain technology. Due to the lack of available real-world datasets, we have generated a synthetic dataset that closely mirrors real-world IoV-MEC-blockchain interactions. This synthetic dataset contains 10,000 records with detailed information such as vehicle IDs, distances traveled, edge node processing capabilities, network delays, and blockchain-specific features. The dataset is designed to reflect real-world patterns, thereby ensuring the relevance and applicability of our findings. You can access the dataset via this link: [https://github.com/Kmoghaddasi/IoV-Synthetic-Dataset].

For the simulation of our model, we created a virtual environment that simulates real-world vehicular networks and edge computing scenarios, taking into account various parameters such as computational resources, server load, data offloading capabilities, and energy consumption levels. The model was trained using our synthetic dataset to develop a robust understanding of the system. We design our simulation environment following the approach outlined in reference [30].

To expedite the validation of our proposed optimization scheme's convergence and efficacy within the vehicular network, and to streamline the network model, we have conducted simulations in both small and large-scale research environments. In this context, the research encompasses a range of 50 to 300 mobile vehicles [47], six BSs [48], and strategically positioned edge computing servers across the study area. Additionally, each BS is intricately linked with an MEC server and is surrounded by three to four RSUs [49]. The computational capacity of the mobile vehicles has been standardized at 1 GHz [50]. Furthermore, we have assessed the computing resource states of MEC servers utilized in vehicle requests and consensus processes, classifying them into four levels: low, medium, high, and very high [48]. Table 2, showcases the utilized simulation environment parameters.

C. TRAINING AND VALIDATION

The training for DDQN, DQN, and DDPG, models was carried out over 1000 episodes, utilizing a learning rate of 0.001 and a batch size of 64. These models were trained using the synthetic dataset, with performance evaluation across the preselected metrics. The training process was replicated thrice for each model to negate randomness in initial weights and the environment. For the random Offloading strategy, offloading decisions were made according to their fundamental principles, thus eliminating the requirement of a training process.

VI. RESULTS AND DISCUSSION

Our proposed model was meticulously tested over 1,000 episodes to examine its learning progression and ability to

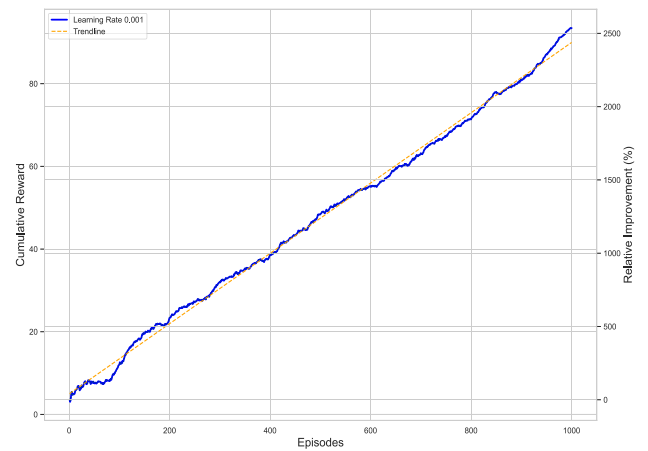
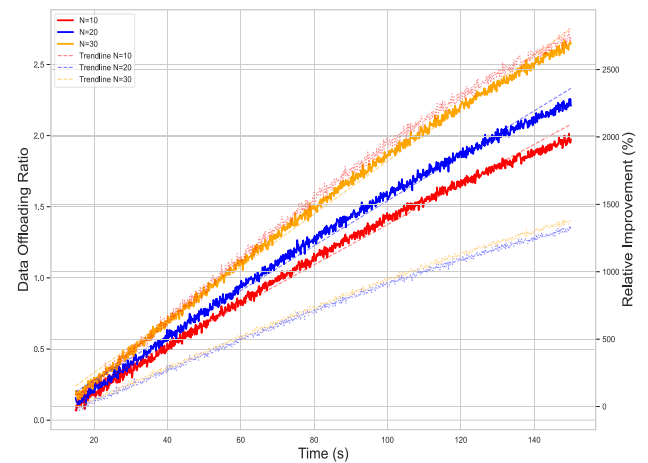
TABLE 2. Simulation environment parameters.

Parameter	Value
Server CPU	Intel i9-7920X
CPU (Cores/Threads)	12/24
CPU Frequency (GHz)	4.3
Memory (GB)	64
Memory Type	DDR4 2400MT/s
Storage (GB)	2048
Storage Type	NVMe SSD
Network Bandwidth (Gbps)	100
GPU	Nvidia RTX 2080 Ti (Dual)
GPU (CUDA Cores)	4352 (per GPU)
GPU Memory (GB)	11 (per GPU)
Virtualization Technology	Virtual Machines
Cooling Solution	Liquid Cooling
Operating System	Ubuntu 18.04
Software Platform	Python 3.8
Deep Learning Library	TensorFlow
Containerization Platform	Docker
Workflow Management	SLURM
Learning Rate	0.001
Batch Size	64
Training Episodes	1000
Count of blockchain nodes B	6 [51]
Computational cycles needed for signature generation and verification α	1 M cycle [49]
Computational cycles needed for MAC generation and verification β	10 M cycle [49]
Mean transaction size during transitions δ	200 B [52]
Maximum cumulative transaction size d	9 M [52]
Average computing capabilities of MEC servers	2.5 GHz [53]

gather rewards, as depicted in Fig. 5. Initially, the model was mainly in the exploration phase, securing a starting reward of about 3.069 units. As the number of episodes increased, a noticeable improvement in the model's skill was observed. This upward trend hit a high point at the 1,000th episode, where the model accomplished a maximum reward of 93.467 units.

The average reward accumulated over the episodes offered additional insights into our DDQN agent's learning journey. by the halfway mark at 500 episodes, the model had significantly enhanced its performance, yielding an average reward of around 47.502 units. The notable rise in reward value, along with a learning rate of 0.001, suggests that the model was successful in interpreting its environment and optimizing its decisions effectively.

The data offloading ratio was thoroughly examined for the different number of vehicle (N10, N20, and N30) configurations over distinct time intervals as shown in Fig. 6. In the N10 configuration, the offloading ratio began at approximately 0.079 at the 15-second mark and steadily increased to about 2.00 by the 150-second mark, marking a substantial increase. The N20 configuration showed a beginning offloading ratio

**FIGURE 5.** Evaluated reward result for learning rate 0.001.**FIGURE 6.** Data offloading ratios for different vehicle counts (N=10, N=20, N=30).

of around 0.093 after 15 seconds and climbed to a high of roughly 2.26 at the 150-second point. The N30 configuration demonstrated the most considerable increase in the offloading ratio, starting from approximately 0.154 after 15 seconds and surging to a peak of about 2.673 after 150 seconds. The progression of data suggests a clear pattern: as the number of vehicles grows from N10 through to N30, there is a notable escalation in the data offloading ratio. This indicates that the capability for offloading data expands with the addition of more vehicles, following a consistent and positive trajectory.

According to Fig. 7, a comprehensive evaluation of the block generation and transaction validation times was conducted at different time intervals. The initial block generation time was recorded to be approximately 121.23 seconds at the 15-second interval. This generation's time saw a considerable reduction, recorded at about 76.41 seconds at the 82.3-second mark. From the 124.6-second mark and onwards, the block generation time remarkably reduced and remained constant at 1.7 seconds, highlighting the model's improved efficiency. Similar to block generation time, the transaction validation time also showcased a decreasing pattern

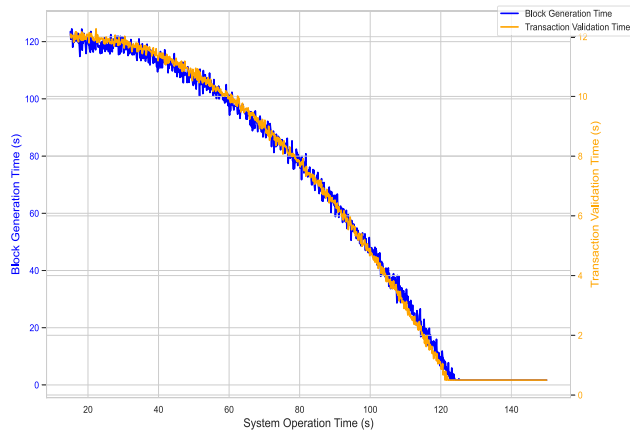


FIGURE 7. Block generation time and transaction validation time across various operational periods.

over time. At the 15-second mark, the validation time was approximately 12.03 seconds. This time decreased to about 7.57 seconds at the 82.3-second mark. Significantly, from the 124.6-second interval to the end of the evaluation period, the validation time mirrored the block generation time, remaining constant at a low 0.5 seconds.

Based on Fig. 8, an analysis was conducted on the energy consumption patterns across networks consisting of 50, 100, and 150 vehicles, considering various algorithms. Focusing on the average energy consumption of the network with 50 vehicles, the DDQN model presented an energy consumption improvement of approximately 26.4% when compared to the next best model, DQN. For networks with 100 vehicles, the percentage improvement in energy consumption rose to approximately 27.7%. Further increase was noted for the network with 150 vehicles, where the improvement was approximately 28.5%. A similar trend was witnessed when total energy consumption was considered. For the 50-vehicle network, the DDQN model displayed an improvement of about 25.5% over the DQN model.

The percentage improvement rose to approximately 26.5% for the network with 100 vehicles, and further increased to approximately 27.5% for the network with 150 vehicles.

Fig. 9, presents six box plots illustrating the energy consumption distribution and total energy consumption for the DDQN, DQN, and DDPG with different vehicle counts. In the network with 50 vehicles (Table 3 & Table 6), the DDQN algorithm performs exceptionally well, having the lowest minimum energy consumption of 15.85 J, nearly 29% less than the DQN's minimum, thereby indicating its higher efficiency. Similarly, the mean energy consumption for the DDQN algorithm is also the lowest, recording 19.60 J, which is approximately 26% less than the DQN's mean. DDQN also outperforms in terms of total energy consumption, with a mean of 548.26 J, roughly 25% lower than DQN's mean. The pattern continues in the 100-vehicle network (Table 4 & Table 7), where the DDQN algorithm maintains its efficiency, achieving a minimum energy consumption of 13.80 J, which

is about 30% lower than the DQN's minimum. DDQN also records the lowest mean energy consumption of 17.24 J, around 28% less than the DQN's mean.

In terms of total energy consumption, DDQN exhibits the best performance with a mean total energy consumption of 960.41 J, approximately 26% lower than the DQN's mean. For the 150-vehicle network (Table 5 & Table 8), the box plot illustrates that the DDQN records the lowest minimum total energy consumption of 12.67 J, about 31% lower than the DQN's minimum. Furthermore, the mean total energy consumption for the DDQN algorithm is also the lowest, measuring 15.92 J, roughly 29% less than the DQN's mean.

The spider plots provide a visual representation of the comparative performance of the four different models in terms of energy consumption as depicted in Fig. 10. They allow us to identify the strengths and weaknesses of each model concerning energy usage in two contexts - average and total energy consumption. Regarding average energy consumption, the DDQN model exhibits the most energy-efficient behavior, with an average energy consumption of 15.93 units. This implies that, on average, the DDQN model consumes the least amount of energy per task offloaded, highlighting its superiority in terms of energy efficiency. Following the DDQN model, the DQN and DDPG models consume comparatively more energy per task, with average energy consumption of 22.32 and 28.74 units, respectively. The random model significantly underperforms in this context, consuming a substantial 84.35 units of energy on average per task. In terms of total energy consumption, the trends from the average energy consumption analysis are consistent. The DDQN model proves to be the most energy-efficient, consuming a total of 2389.40 units for all tasks offloaded. On the other hand, the DQN and DDPG models exhibit higher total energy consumption at 3347.77 and 4311.35 units, respectively. Again, the random model shows the highest energy consumption, with a total energy usage of 12652.56 units.

Based on the analysis demonstrated in Fig. 11, we investigated the energy efficiency, measured in terms of IoV per Joule, across networks with 50, 100, and 150 vehicles utilizing various algorithms. Specifically, for the network with 50 vehicles, the DDQN model showed an improvement in average energy efficiency of approximately 33% compared to the DQN model, which had the second-best performance. When the network size increased to 100 vehicles, the DDQN model maintained its superior performance, yielding an improvement in energy efficiency of approximately 35.5% compared to the DQN model. This trend continued in the network with 150 vehicles, where the improvement in energy efficiency reached approximately 37.4% compared to the DQN model.

Fig. 12, displays three box plots representing the energy efficiency of the different algorithms in networks with varying vehicle numbers. In the 50-vehicle network (Table 9), the DDQN algorithm proves its efficiency by achieving the highest maximum energy efficiency of 0.049720691 IoV/J, about 35% higher than the DQN's maximum. Similarly, the mean

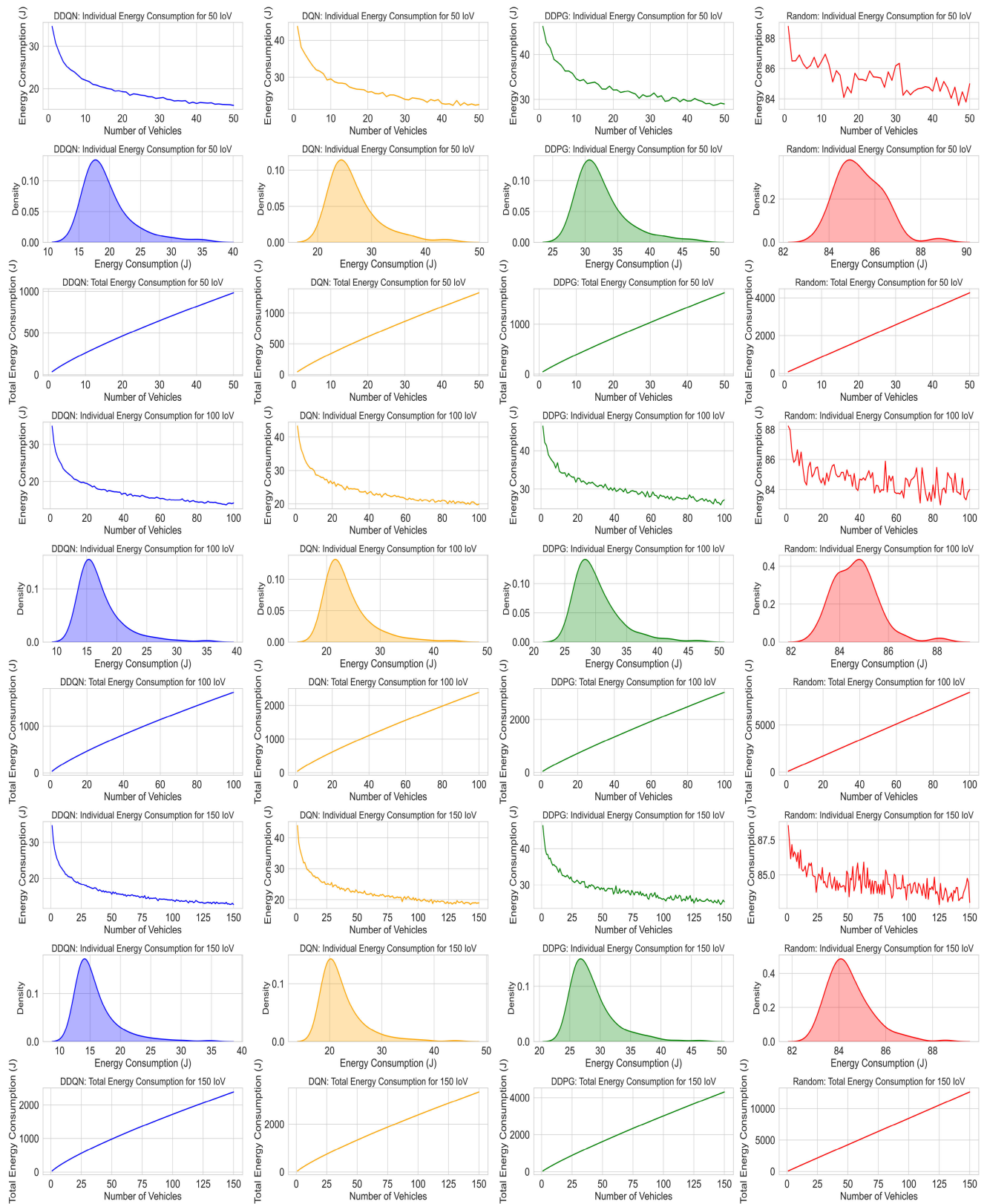


FIGURE 8. The results of energy consumption and total energy consumption of the proposed algorithm and comparison models in different scenarios.

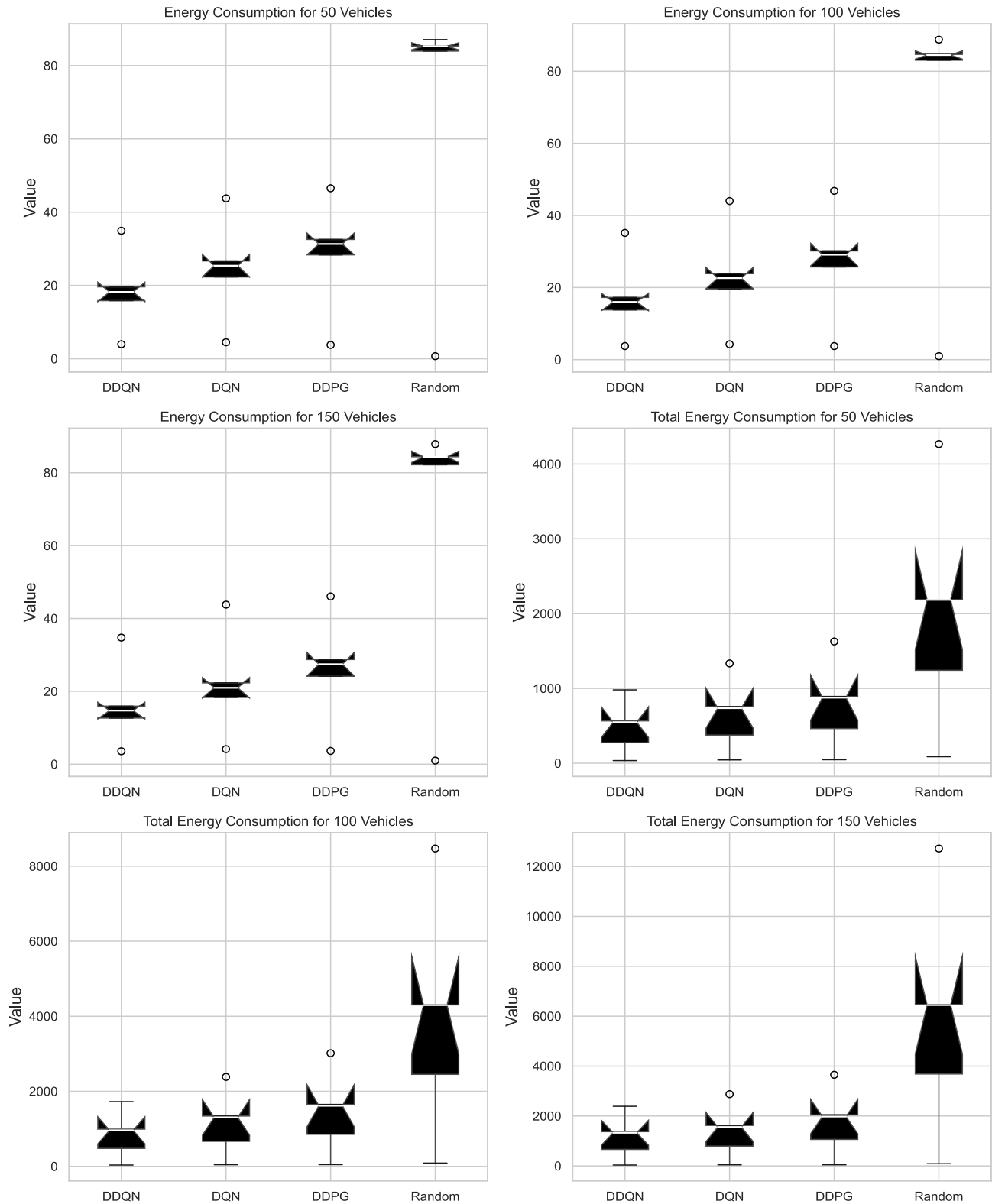


FIGURE 9. Box plot visualization showcasing the distribution and statistical analysis of individual and total energy consumption with different numbers of vehicles.

energy efficiency for the DDQN algorithm is also the highest, recording 0.040264942 IoV/J, which is approximately 33%

more than the DQN’s mean. The median energy efficiency is also highest for DDQN, at 0.043085507 IoV/J, about 33%

TABLE 3. The statistical analysis of energy consumption (J) for the algorithms in a network with 50 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	15.85302859	22.31042096	28.3412625	83.94346122
Max	34.91500673	43.76052086	46.51222661	87.07729067
Mean	19.6036098	26.67771352	32.55005742	85.33832797
50%	18.218061	25.37550882	31.31202511	85.25206835
Std.	3.957620424	4.486452821	3.759393959	0.704250221

TABLE 4. The statistical analysis of energy consumption (J) for the algorithms in a network with 100 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	13.80110545	19.62448524	25.70502179	83.07131712
Max	35.15268891	44.00203206	46.81723549	88.78289998
Mean	17.24574919	23.83181688	30.15196024	84.70193849
50%	15.99551036	22.6061236	29.1015843	84.51223317
Std.	3.753873672	4.253623947	3.748666249	0.962634515

TABLE 5. The statistical analysis of energy consumption (J) for the algorithms in a network with 150 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	12.67412107	18.33693502	24.1403801	82.17887095
Max	34.74423972	43.79406788	46.04656503	87.85908991
Mean	15.92889997	22.31337838	28.71935553	84.39048346
50%	14.71166819	20.97218366	27.43651804	84.35584562
Std.	3.556386416	4.143799351	3.640917304	0.992800489

TABLE 6. The statistical analysis of total energy consumption (J) for the algorithms in a network with 50 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	34.91500673	43.76052086	46.51222661	86.79805332
Max	980.1804898	1333.885676	1627.502871	4266.916399
Mean	548.2618538	735.7325186	875.9290754	2183.455208
50%	565.5996724	756.1682527	891.8486078	2186.463089
Std.	274.1953887	376.0953461	463.29213	1242.587053

TABLE 7. The statistical analysis of total energy consumption (J) for the algorithms in a network with 100 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	35.15268891	44.00203206	46.81723549	87.74655036
Max	1724.574919	2383.181688	3015.196024	8470.193849
Mean	960.4180465	1305.568327	1614.626443	4297.202815
50%	991.9109406	1341.414808	1649.927644	4303.331922
Std.	479.4557832	668.1515185	855.2368731	2452.497835

TABLE 8. The statistical analysis of total energy consumption (J) for the algorithms in a network with 150 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	34.74423972	43.79406788	46.04656503	87.75010218
Max	2389.334996	3347.006757	4307.903329	12658.57252
Mean	1328.11331	1832.336863	2301.913266	6401.908112
50%	1372.116981	1882.922274	2351.54938	6412.120017
Std.	662.9028735	935.8725389	1220.497384	3661.673768

higher than the DQN's median. This trend continues in the 100-vehicle network (Table 10), where the DDQN algorithm maintains its superior energy efficiency, achieving a max-

imum of 0.057508308 IoV/J, which is about 39% higher than the DQN's maximum. DDQN also records the highest mean energy efficiency of 0.047213049 IoV/J, around 35%

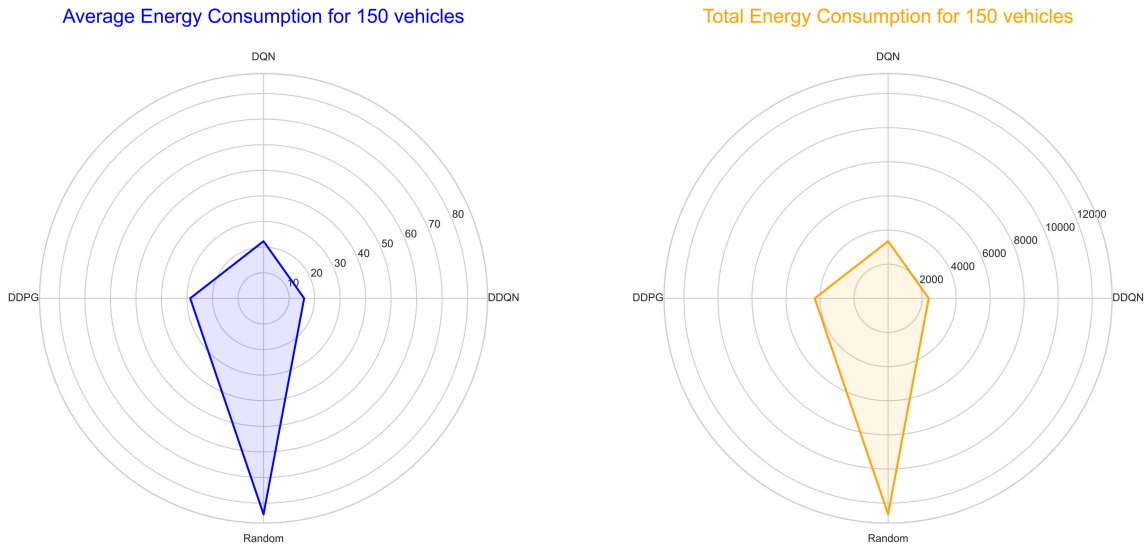


FIGURE 10. Spider plot illustrating the comparative analysis of the average and total energy consumption of 150 vehicles.

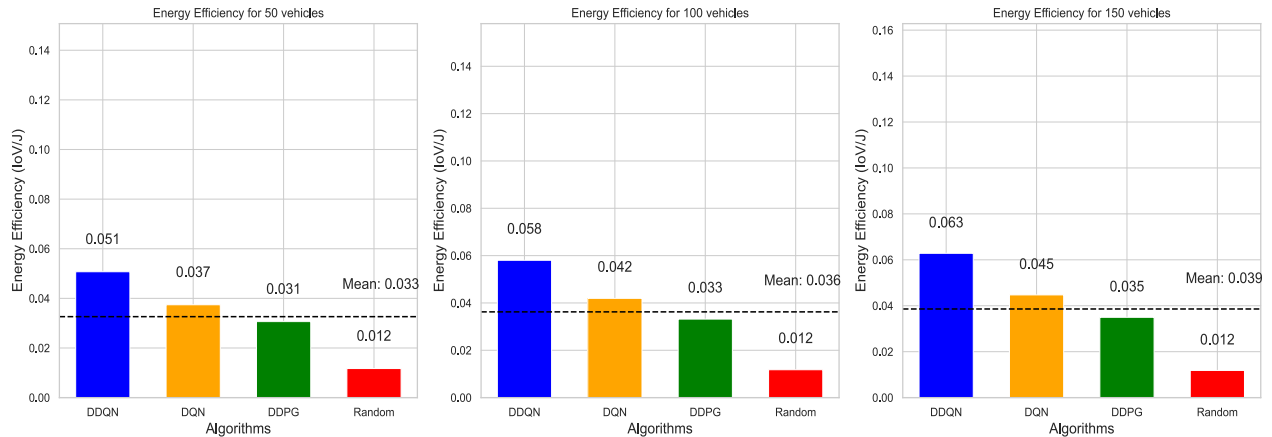


FIGURE 11. The result of energy efficiency of the proposed algorithm and comparison models in different scenarios.

more than the DQN's mean. For the 150-vehicle network (Table 11), DDQN retains its superiority. It records the highest maximum energy efficiency of 0.062349271 IoV/J, about 40% higher than the DQN's maximum. Furthermore, the mean energy efficiency for the DDQN algorithm is also the highest, measuring 0.051541344 IoV/J, roughly 37% more than the DQN's mean.

In Fig. 13, the latency patterns across networks with 150 vehicles for various models are evaluated. The latency, measured in milliseconds, is a crucial factor in the performance of the network, where lower latency indicates higher efficiency. In the network with 150 vehicles, the DDQN model's average latency was approximately 28.42% lower than that of the other best algorithm.

Fig. 14, displays three box plots indicating the latency for different models across networks with different numbers of vehicles. In a network with 50 vehicles (Table 12), the DDQN algorithm's latency rates prove to be comparatively

better, with the lowest maximum latency of 106.7387004ms, nearly 35% less than the maximum latency of the DDPG algorithm. Further, the DDQN's mean latency was also the least at 87.76748916ms, which is approximately 28% lower than the random algorithm's mean. In the 100-vehicle network scenario (Table 13), the DDQN algorithm continues to outperform the others with the lowest maximum latency at 119.9032207ms, roughly 60% lower than the DDPG algorithm's maximum. The mean latency for the DDQN algorithm was also the lowest at 90.34247933ms, approximately 77% less than the DDPG's mean. In the 150-vehicle network (Table 14), the DDQN shows the lowest maximum latency at 156.5937039ms, about 51% lower than the DDPG algorithm's maximum. DDQN also had the lowest mean latency of 105.6249756ms, roughly 45% less than the DDPG's mean.

The cost-effectiveness of the DDQN, DQN, DDPG, and random models across networks with 50, 100, and 150 vehi-

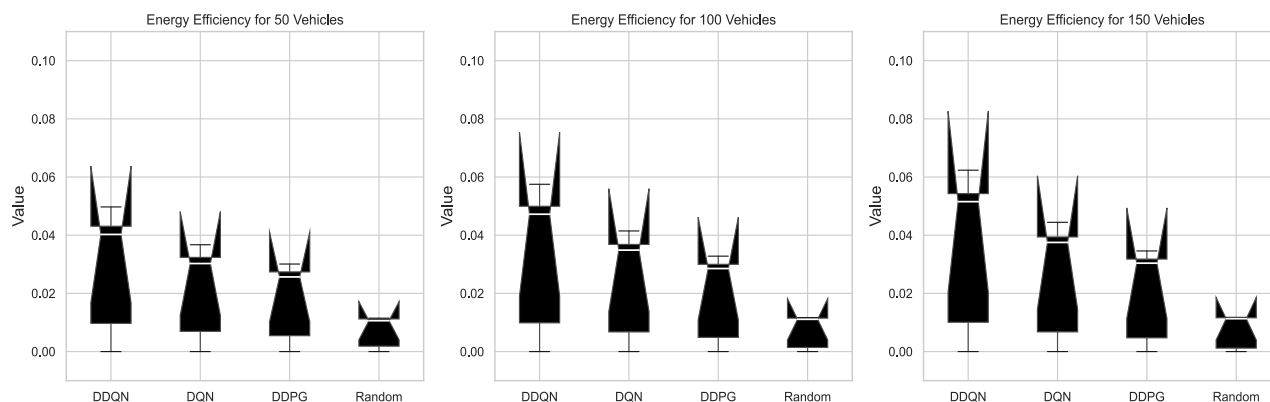


FIGURE 12. Box plot representation of energy efficiency distribution and statistical insights with different numbers of vehicles.

TABLE 9. The statistical analysis of energy efficiency (IoV/I) for the algorithms in a network with 100 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	-	-	-	-
Max	0.049720691	0.03672532	0.030089233	0.011485337
Mean	0.040264942	0.030264427	0.025656725	0.010610134
50%	0.043085507	0.032380588	0.027386269	0.011212447
Std.	0.009689437	0.006951648	0.005456241	0.001859619

TABLE 10. The statistical analysis of energy efficiency (IoV/I) for the algorithms in a network with 150 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	-	-	-	-
Max	0.057508308	0.04146657	0.032798644	0.011690417
Mean	0.047213049	0.034872784	0.028545588	0.011091399
50%	0.049951696	0.036835292	0.029996053	0.0114964
Std.	0.009872006	0.006759761	0.004879042	0.001413397

TABLE 11. The statistical analysis of energy efficiency (IoV/I) for the algorithms in a network with 50 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	-	-	-	-
Max	0.062349271	0.044430167	0.034589542	0.01176221
Mean	0.051541344	0.037514605	0.030387079	0.011304002
50%	0.054301304	0.039393635	0.031791864	0.011612241
Std.	0.010075756	0.006770604	0.004716958	0.00118666

TABLE 12. The statistical analysis of latency (ms) for the algorithms in a network with 50 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	69.01134934	57.6379509	2.340199536	337.2449377
Max	106.7387004	164.8310073	170.9682728	488.3947016
Mean	87.76748916	94.25623131	90.73648697	399.152053
50%	86.33484771	94.997734	92.61988396	395.7700513
Std.	9.247048828	18.55332392	33.22595775	37.388746

cles is discussed in Fig. 15. In the 50-vehicle network, the DDQN model proved to be the most efficient, with an average cost that was approximately 7.41% less than the DQN model, the second-best performer. In the 100-vehicle network, the

DDQN model showed an improvement in cost efficiency of approximately 18.28% compared to the DQN model. Furthermore, in the highest density scenario with 150 vehicles, the DDQN model had an average cost that was approximately

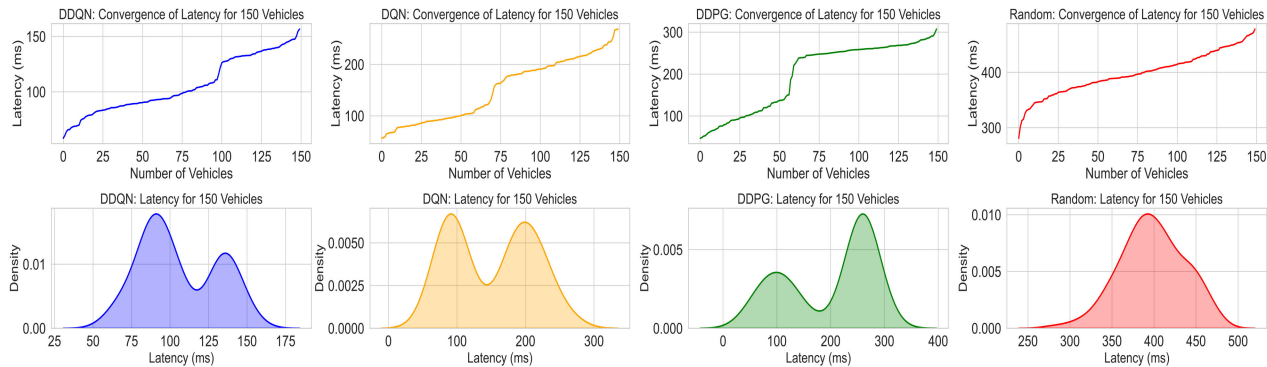


FIGURE 13. The result of latency of the proposed algorithm and comparison models in a network with 150 vehicles.

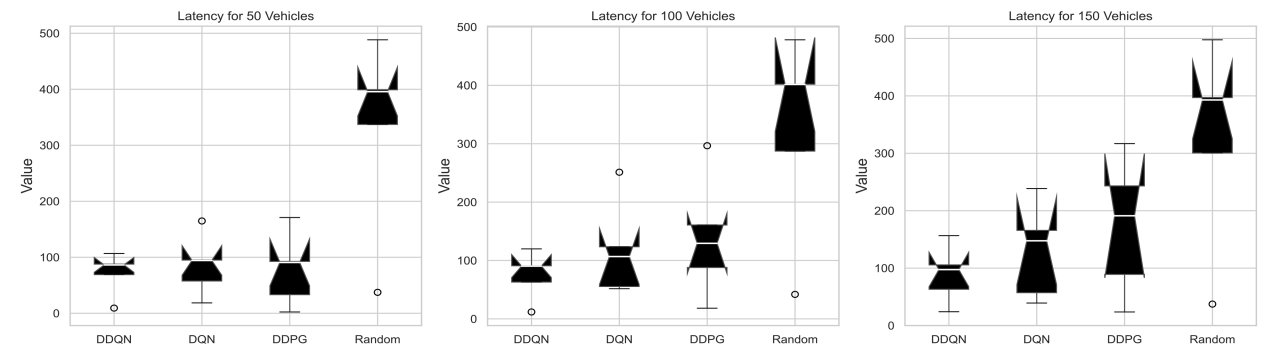


FIGURE 14. Box plot analysis illustrating the distribution and key statistical measures of network latency across various numbers of vehicles.

TABLE 13. The statistical analysis of latency (ms) for the algorithms in a network with 100 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	63.04104514	55.43972184	18.25031359	287.3817842
Max	119.9032207	251.2236206	296.5868675	478.0427344
Mean	90.34247933	123.5235425	160.7801642	401.5514982
50%	90.97716414	106.7686741	129.3870673	401.969816
Std.	11.83705954	51.73075062	87.98226342	41.94580657

TABLE 14. The statistical analysis of latency (ms) for the algorithms in a network with 150 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	63.17253595	39.18321544	23.6509555	300.588432
Max	156.5937039	238.6395366	316.8611556	497.657839
Mean	105.6249756	147.6081514	191.202904	397.0286948
50%	97.46488954	165.7637164	243.1821533	392.8981827
Std.	24.08665704	57.20015983	89.1042797	37.36114098

23.75% lower than the DQN model. Regarding the total cost, in the 50-vehicle network, the proposed model demonstrated an average total cost that was approximately 10.74% less than the DQN model. In the 100 and 150-vehicle network, the DDQN model showed an improvement in total cost efficiency by about 19.56% and 23.77% lower than that of the DQN model, respectively.

Fig. 16, displays six box plots representing the cost and total cost of the different algorithms in networks with

50, 100, and 150 vehicles respectively. For the 50-vehicle network (Table 15 & Table 18), the DDQN algorithm demonstrates the most cost-efficient performance with the lowest maximum cost of 6156.433779 and a total cost of 6236.05746. Also, DDQN showcases a lower mean cost of 3125.364797 and a total cost of 3137.057113. Notably, DDQN outperforms the other algorithms with a lower median cost of 3143.496773 and a total cost of 3141.414619. In a network with 100 vehicles (Table 16 &

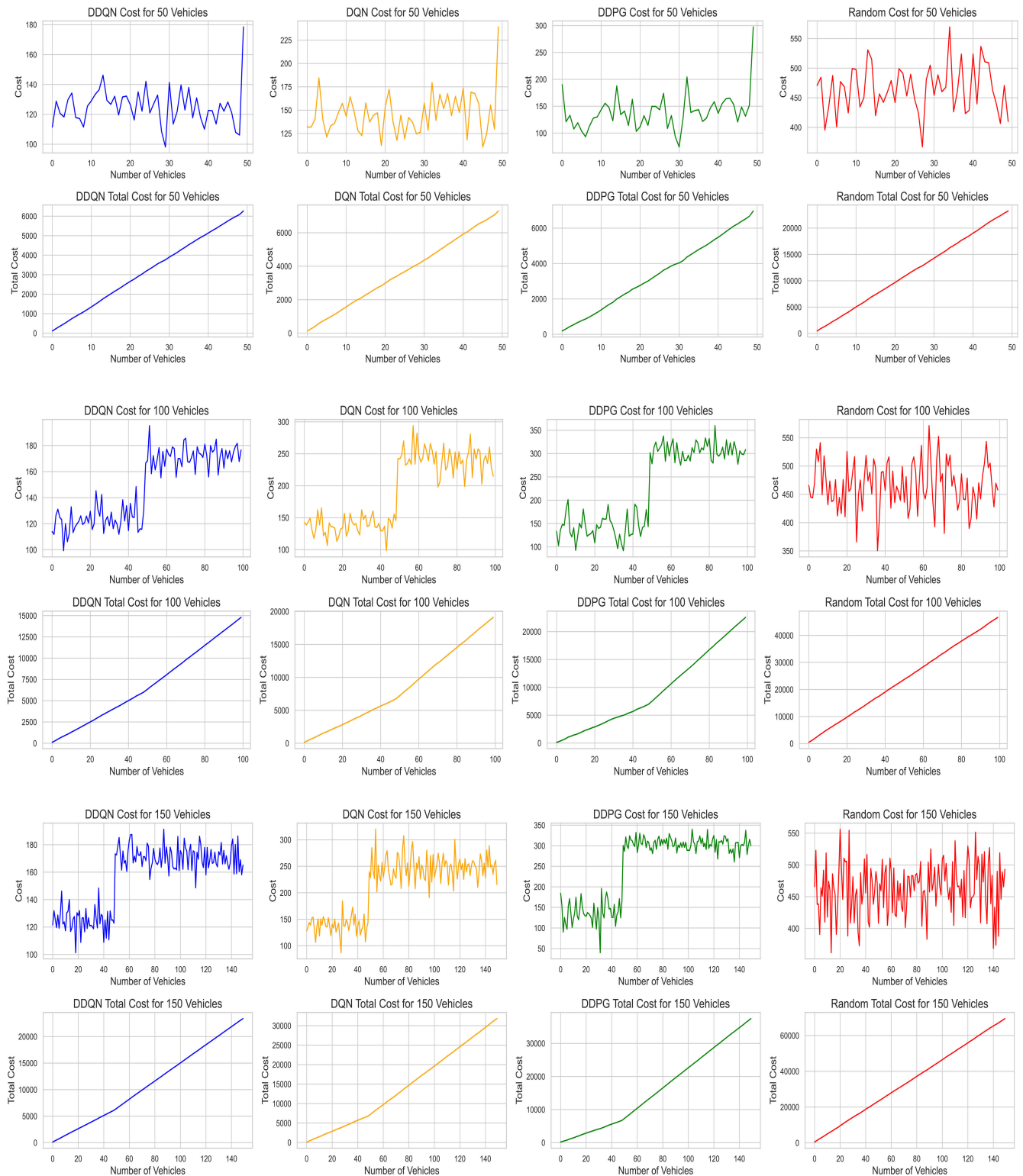


FIGURE 15. The result of cost and total cost of the proposed algorithm and comparison models in different scenarios.

Table 19), the DDQN algorithm again shows better cost efficiency with the lowest maximum cost of 14820.82781 and a total cost of 14846.50835. Moreover, the mean cost

for the DDQN algorithm was 6883.529495, and the total cost was 6895.304707, both being the lowest among all the algorithms. In the case of the 150-vehicle network

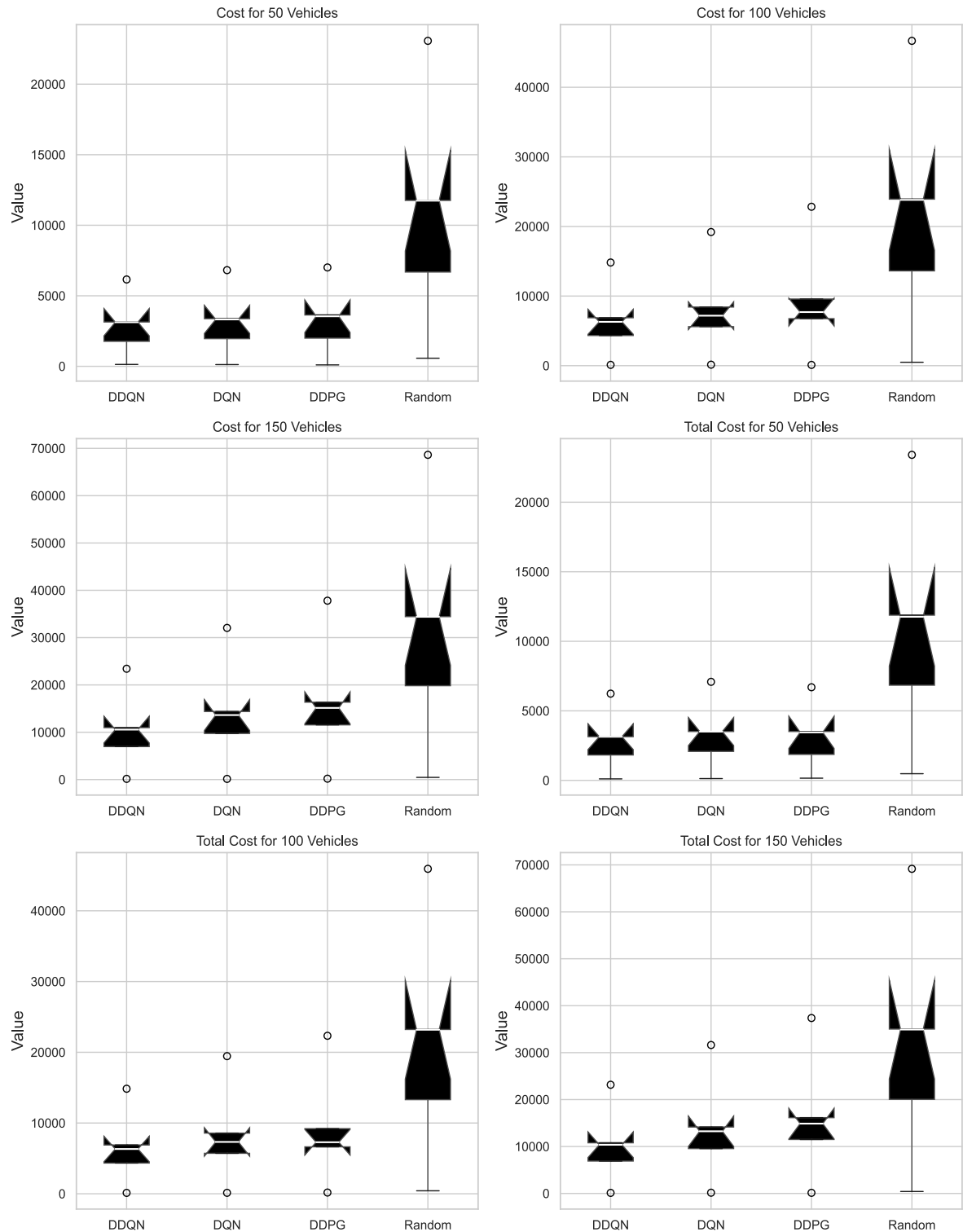


FIGURE 16. Box plot visualization showcasing the distribution and significant statistical parameters of individual and total costs throughout different numbers of vehicles.

(Table 17 & Table 20), the DDQN algorithm continued to display superior cost-efficient performance, recording the lowest maximum cost of 23422.20072 and a total cost of 23154.84507. Additionally, the DDQN algorithm showed

the lowest mean cost of 10972.27826 and a total cost of 10788.36926.

The correlation chart presents a comprehensive picture of the interrelationship among various metrics of the DDQN,

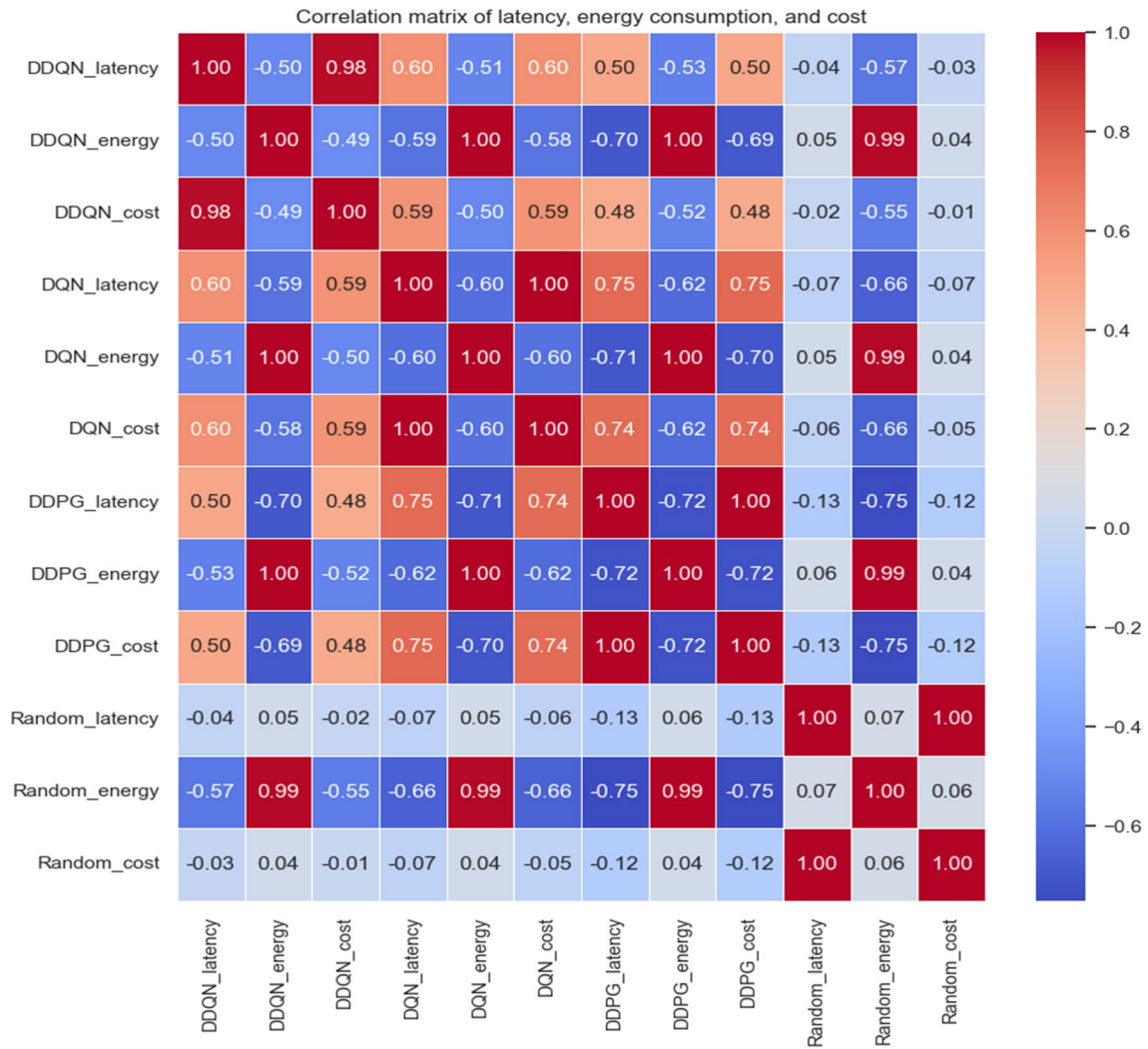


FIGURE 17. Comprehensive analysis of the correlation among various metrics of the DDQN, DQN, DDPG, and random offloading models.

TABLE 15. The statistical analysis of cost for the algorithms in a network with 50 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	133.522891	122.9039082	101.0951405	574.6757723
Max	6156.433779	6817.277893	7008.452246	23070.04995
Mean	3125.364797	3375.813762	3547.39437	11758.95877
50%	3143.496773	3323.879562	3639.26671	11741.78932
Std.	1770.493352	1960.196173	1998.735672	6683.572735

TABLE 16. The statistical analysis of cost for the algorithms in a network with 100 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	115.2472651	140.6703219	110.6436768	488.4563588
Max	14820.82781	19201.35406	22831.43977	46669.06858
Mean	6883.529495	8424.115806	9575.158741	23819.88383
50%	6298.904494	7182.523013	7700.123076	23925.65964
Std.	4331.476192	5605.332566	6753.640387	13618.72007

DQN, DDPG, and random models, as illustrated in Fig. 17, In correlation analysis, the correlation coefficient ranges between -1 and 1. A value closer to 1 signifies a high pos-

itive correlation, whereas a value nearer to -1 indicates a strong negative correlation. A value approximating zero indicates little to no correlation. Focusing on the DDQN

TABLE 17. The statistical analysis of cost for the algorithms in a network with 150 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	148.1352011	140.6424951	185.5136428	454.8348007
Max	23422.20072	32049.46242	37807.42145	68631.36507
Mean	10972.27826	14387.10549	16349.55585	34431.95018
50%	10556.3315	13630.16142	15142.898	34394.80649
Std.	6962.679499	9746.036876	11552.84374	19830.80632

TABLE 18. The statistical analysis of total cost for the algorithms in a network with 50 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	107.0612658	128.7224188	158.3743151	478.8288443
Max	6236.05746	7085.626408	6695.661237	23406.78276
Mean	3137.057113	3514.295582	3451.295728	11878.53322
50%	3141.414619	3508.161219	3514.235973	11762.83025
Std.	1824.886723	2082.653203	1865.865411	6841.105776

TABLE 19. The statistical analysis of total cost for the algorithms in a network with 100 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	127.4892403	134.0492077	173.5430128	422.2133595
Max	14846.50835	19452.80969	22332.25935	45931.20305
Mean	6895.304707	8573.350069	9207.626147	23222.93313
50%	6327.287893	7328.406998	7286.841448	23186.48551
Std.	4342.213194	5719.551717	6635.933026	13303.02791

TABLE 20. The statistical analysis of total cost for the algorithms in a network with 150 vehicles.

	DDQN	DQN	DDPG	Random Offloading
Min	122.8089262	157.257134	132.7245429	414.1494577
Max	23154.84507	31624.74677	37374.53267	69159.4336
Mean	10788.36926	14155.28161	16146.5673	34921.24842
50%	10371.13269	13244.51064	14876.76085	35023.33646
Std.	6898.848331	9554.354762	11498.24653	20047.8222

model initially, the correlation coefficient between latency and cost is exceedingly strong at 0.977. This suggests a direct and robust relationship where an increase in latency corresponds with a rise in the cost metric. This correlation implies that any task offloading strategy that successfully minimizes latency would be simultaneously effective in reducing the cost, a critical consideration in the performance optimization of IoV-MEC systems. Interestingly, a counteractive relationship is observed in the DDQN model between latency and energy, with a correlation coefficient of -0.506. This inverse correlation signifies that as the latency metric decreases, there is a corresponding increase in energy consumption and vice versa. This result underscores a prevalent challenge in these systems: managing the trade-off between optimizing latency (and, by extension, cost) and maintaining energy efficiency. Observations for the DQN model are parallel to those noted in the DDQN model. A high positive correlation of 0.996 between latency and cost indicates that these two metrics are intrinsically linked, suggesting that cost optimization strategies in IoV-MEC systems could significantly benefit from a focus on latency reduction. Furthermore, the correlation between energy and latency remains nega-

tive (-0.622), further emphasizing the opposing relationship between energy efficiency and latency optimization. Similar patterns of correlations are observed in the DDPG model, where the correlation between latency and cost is 0.998. Again, this suggests that strategies designed to decrease latency could have the secondary effect of cost reduction. The energy-latency correlation remains negative at -0.691, indicating a persistent challenge in balancing these two critical metrics. Finally, the random model correlation analysis reveals an extremely high positive correlation (0.994) between latency and cost, reinforcing the pattern seen across all models. The energy-latency correlation remains negative (-0.159), once again affirming the inverse relationship between these two parameters across all models.

VII. CONCLUSION AND FUTURE WORKS

In this paper, we introduce a multi-objective task offloading strategy that integrates the DDQN algorithm with blockchain technology, aiming to enhance the performance of IoV systems integrated with MEC. This approach takes into account multiple objectives, enabling a sophisticated trade-off analysis to effectively balance and optimize overall system

performance. Our model demonstrates superior performance in several aspects, improving energy efficiency by up to 26.4%, reducing latency by 6.87%, and curbing costs by 7.41%. However, our model does have some limitations. Its primary reliance on single-hop communication might restrict its effectiveness when handling larger and more complex networks. Also, its theoretical foundation and simulation-based nature might limit its practical applicability and real-world performance.

In future work, we aim to refine and expand our model, evolving it to support multi-hop communication for enhanced utility in complex network scenarios. Real-world testing is essential to ensure the robustness and reliability of our model beyond controlled simulations. Comprehensive assessments under diverse network conditions, including high vehicle mobility and fluctuating traffic, will be conducted to enhance adaptability and resilience. Additionally, addressing potential adversarial attacks on the security of our Blockchain-enabled framework is crucial for real-world applicability, and exploring methods to mitigate challenges associated with varying network densities, especially in scenarios with sporadic connectivity or congestion, will contribute to the versatility and reliability of our proposed model.

REFERENCES

- [1] P. K. R. Maddikunta, Q.-V. Pham, D. C. Nguyen, T. Huynh-The, O. Aouedi, G. Yenduri, S. Bhattacharya, and T. R. Gadekallu, "Incentive techniques for the Internet of Things: A survey," *J. Netw. Comput. Appl.*, vol. 206, Oct. 2022, Art. no. 103464.
- [2] M. Soori, B. Arezoo, and R. Dastres, "Internet of Things for smart factories in Industry 4.0, a review," *Internet Things Cyber-Phys. Syst.*, vol. 3, pp. 192–204, Apr. 2023.
- [3] K. Wang, X. Wang, and X. Liu, "A high reliable computing offloading strategy using deep reinforcement learning for IoTs in edge computing," *J. Grid Comput.*, vol. 19, no. 2, pp. 1–15, Jun. 2021.
- [4] S. El Madani, S. Motahhir, and A. El Ghzizal, "Internet of Vehicles: Concept, process, security aspects and solutions," *Multimedia Tools Appl.*, vol. 81, no. 12, pp. 16563–16587, May 2022.
- [5] B. Hildebrand, M. Baza, T. Salman, S. Tabassum, B. Konatham, F. Amsaad, and A. Razaque, "A comprehensive review on blockchains for Internet of Vehicles: Challenges and directions," *Comput. Sci. Rev.*, vol. 48, May 2023, Art. no. 100547.
- [6] H. Pourrahmani, A. Yavarinasab, R. Zahedi, A. Gharehghani, M. H. Mohammadi, P. Bastani, and J. Van Herle, "The applications of Internet of Things in the automotive industry: A review of the batteries, fuel cells, and engines," *Internet Things*, vol. 19, Aug. 2022, Art. no. 100579.
- [7] H. Taslimasa, S. Dadkhah, E. C. P. Neto, P. Xiong, S. Ray, and A. A. Ghorbani, "Security issues in Internet of Vehicles (IoV): A comprehensive survey," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100809.
- [8] K. Shah, S. Chadotra, S. Tanwar, R. Gupta, and N. Kumar, "Blockchain for IoV in 6G environment: Review solutions and challenges," *Cluster Comput.*, vol. 25, no. 3, pp. 1927–1955, Jun. 2022.
- [9] F. Ajaz, M. Naseem, G. Ahamad, Q. R. Khan, S. Sharma, and E. Abbasi, "Routing protocols for Internet of Vehicles: A review," in *AI and Machine Learning Paradigms for Health Monitoring System*. 2021, pp. 95–103. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-33-4412-9_5
- [10] K. Moghaddasi and S. Rajabi, "Double deep Q-learning networks for energy-efficient IoT task offloading in D2D MEC environments," in *Proc. 7th Int. Conf. Internet Things Appl. (IoT)*, Oct. 2023, pp. 1–6, doi: 10.1109/IoT60973.2023.10365356.
- [11] K. Moghaddasi and M. Masdari, "Blockchain-driven optimization of IoT in mobile edge computing environment with deep reinforcement learning and multi-criteria decision-making techniques," *Cluster Comput.*, vol. 26, pp. 1–29, Nov. 2023. [Online]. Available: <https://link.springer.com/article/10.1007/s10586-023-04195-4>, doi: 10.1007/s10586-023-04195-4.
- [12] C. Feng, P. Han, X. Zhang, B. Yang, Y. Liu, and L. Guo, "Computation offloading in mobile edge computing networks: A survey," *J. Netw. Comput. Appl.*, vol. 202, Jun. 2022, Art. no. 103366. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1084804522000327>
- [13] K. Moghaddasi and S. Rajabi, "Learning at the edge: Mobile edge computing and reinforcement learning for enhanced web application performance," in *Proc. 9th Int. Conf. Web Res. (ICWR)*, May 2023, pp. 300–304.
- [14] T. Li, X. He, S. Jiang, and J. Liu, "A survey of privacy-preserving offloading methods in mobile-edge computing," *J. Netw. Comput. Appl.*, vol. 203, Jul. 2022, Art. no. 103395.
- [15] Q. Tang, H. Lyu, G. Han, J. Wang, and K. Wang, "Partial offloading strategy for mobile edge computing considering mixed overhead of time and energy," *Neural Comput. Appl.*, vol. 32, no. 19, pp. 15383–15397, Oct. 2020.
- [16] A. Mohammed, H. Nahom, A. Tewodros, Y. Habtamu, and G. Hayelom, "Deep reinforcement learning for computation offloading and resource allocation in blockchain-based multi-UAV-enabled mobile edge computing," in *Proc. 17th Int. Comput. Conf. Wavelet Act. Media Technol. Inf. Process. (ICCWAMTIP)*, Dec. 2020, pp. 295–299.
- [17] A. G. Gad, D. T. Mosa, L. Abualigah, and A. A. Abohany, "Emerging trends in blockchain technology and applications: A review and outlook," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 9, pp. 6719–6742, Oct. 2022.
- [18] J. Chen, J. Wu, H. Liang, S. Mumtaz, J. Li, K. Konstantin, A. K. Bashir, and R. Nawaz, "Collaborative trust blockchain based unbiased control transfer mechanism for industrial automation," *IEEE Trans. Ind. Appl.*, vol. 56, no. 4, pp. 4478–4488, Jul. 2020.
- [19] Y. Chen, Y. Lu, L. Bulysheva, and M. Y. Kataev, "Applications of blockchain in industry 4.0: A review," *Inf. Syst. Frontiers*, pp. 1–15, Feb. 2022. [Online]. Available: <https://link.springer.com/article/10.1007/s10796-022-10248-7>
- [20] A. M. Seid, J. Lu, H. N. Abishu, and T. A. Ayall, "Blockchain-enabled task offloading with energy harvesting in multi-UAV-assisted IoT networks: A multi-agent DRL approach," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 12, pp. 3517–3532, Dec. 2022.
- [21] A. Di Vaio, R. Hassan, and R. Palladino, "Blockchain technology and gender equality: A systematic literature review," *Int. J. Inf. Manag.*, vol. 68, Feb. 2023, Art. no. 102517.
- [22] Z. Ullah, M. Naeem, A. Coronato, P. Ribino, and G. De Pietro, "Blockchain applications in sustainable smart cities," *Sustain. Cities Soc.*, vol. 97, Oct. 2023, Art. no. 104697.
- [23] G. Manogaran, S. Mumtaz, C. X. Mavromoustakis, E. Pallis, and G. Mastorakis, "Artificial intelligence and blockchain-assisted offloading approach for data availability maximization in edge nodes," *IEEE Trans. Veh. Technol.*, vol. 70, no. 3, pp. 2404–2412, Mar. 2021.
- [24] Y. Liu, L. Pan, and S. Chen, "A hierarchical blockchain-enabled security-threat assessment architecture for IoV," *Digit. Commun. Netw.*, Feb. 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352864822002875>
- [25] Y. Xiao, Y. Liu, and T. Li, "Edge computing and blockchain for quick fake news detection in IoV," *Sensors*, vol. 20, no. 16, p. 4360, Aug. 2020.
- [26] Y. Zhang, L. Zhang, Q. Wu, and Y. Mu, "Blockchain-enabled efficient distributed attribute-based access control framework with privacy-preserving in IoV," *J. King Saud Univ.-Comput. Inf. Sci.*, vol. 34, no. 10, pp. 9216–9227, Nov. 2022.
- [27] Q. Li, W. Su, P. Zhang, X. Cheng, M. Li, and Y. Liu, "Blockchain-based method for pre-authentication and handover authentication of IoV vehicles," *Electronics*, vol. 12, no. 1, p. 139, Dec. 2022.
- [28] P. K. Lahiri, D. Das, W. Mansoor, S. Banerjee, and P. Chatterjee, "A trustworthy blockchain based framework for impregnable IoV in edge computing," in *Proc. IEEE 17th Int. Conf. Mobile Ad Hoc Sensor Syst. (MASS)*, Dec. 2020, pp. 26–31.
- [29] D. Zhang, F. R. Yu, and R. Yang, "Blockchain-based multi-access edge computing for future vehicular networks: A deep compressed neural network approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 12161–12175, Aug. 2022.
- [30] X. Ye, M. Li, P. Si, R. Yang, Z. Wang, and Y. Zhang, "Collaborative and intelligent resource optimization for computing and caching in IoV with blockchain and MEC using A3C approach," *IEEE Trans. Veh. Technol.*, vol. 72, no. 2, pp. 1449–1463, Feb. 2023.
- [31] Q. Mei, H. Xiong, Y. Zhao, and K.-H. Yeh, "Toward blockchain-enabled IoV with edge computing: Efficient and privacy-preserving vehicular communication and dynamic updating," in *Proc. IEEE Conf. Dependable Secure Comput. (DSC)*, Jan. 2021, pp. 1–8.

- [32] L. Cui, Z. Chen, S. Yang, Z. Ming, Q. Li, Y. Zhou, S. Chen, and Q. Lu, "A blockchain-based containerized edge computing platform for the Internet of Vehicles," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 2395–2408, Feb. 2021.
- [33] H. Liao, Y. Mu, Z. Zhou, M. Sun, Z. Wang, and C. Pan, "Blockchain and learning-based secure and intelligent task offloading for vehicular fog computing," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4051–4063, Jul. 2021.
- [34] S. Iqbal, A. W. Malik, A. U. Rahman, and R. M. Noor, "Blockchain-based reputation management for task offloading in micro-level vehicular fog network," *IEEE Access*, vol. 8, pp. 52968–52980, 2020.
- [35] X. Zheng, M. Li, Y. Chen, J. Guo, M. Alam, and W. Hu, "Blockchain-based secure computation offloading in vehicular networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4073–4087, Jul. 2021.
- [36] B. K. Sovacool, J. Kester, L. Noel, and G. Zarazua de Rubens, "Actors, business models, and innovation activity systems for vehicle-to-grid (V2G) technology: A comprehensive review," *Renew. Sustain. Energy Rev.*, vol. 131, Oct. 2020, Art. no. 109963.
- [37] A. Kabil, K. Rabieh, F. Kaleem, and M. A. Azer, "Vehicle to pedestrian systems: Survey, challenges and recent trends," *IEEE Access*, vol. 10, pp. 123981–123994, 2022.
- [38] A. W. Thompson and Y. Perez, "Vehicle-to-everything (V2X) energy services, value streams, and regulatory policy implications," *Energy Policy*, vol. 137, Feb. 2020, Art. no. 111136.
- [39] S. Zeadally, J. Guerrero, and J. Contreras, "A tutorial survey on vehicle-to-vehicle communications," *Telecommun. Syst.*, vol. 73, no. 3, pp. 469–489, Mar. 2020.
- [40] A. R. Khan, M. F. Jamlos, N. Osman, M. I. Ishak, F. Dzaharudin, Y. K. Yeow, and K. A. Khairi, "DSRC technology in vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) IoT system for intelligent transportation system (ITS): A review," in *Recent Trends in Mechatronics Towards Industry 4.0*. 2022, pp. 97–106. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-33-4597-3_10
- [41] H. N. Abishu, A. M. Seid, Y. H. Yacob, T. Ayall, G. Sun, and G. Liu, "Consensus mechanism for blockchain-enabled vehicle-to-vehicle energy trading in the Internet of Electric Vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 1, pp. 946–960, Jan. 2022.
- [42] L. Liao, Y. Lai, F. Yang, and W. Zeng, "Online computation offloading with double reinforcement learning algorithm in mobile edge computing," *J. Parallel Distrib. Comput.*, vol. 171, pp. 28–39, Jan. 2023.
- [43] Y. Ju, Y. Chen, Z. Cao, L. Liu, Q. Pei, M. Xiao, K. Ota, M. Dong, and V. C. M. Leung, "Joint secure offloading and resource allocation for vehicular edge computing network: A multi-agent deep reinforcement learning approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 5555–5569, May 2023, doi: [10.1109/TITS.2023.3242997](https://doi.org/10.1109/TITS.2023.3242997).
- [44] H. Tang, H. Wu, G. Qu, and R. Li, "Double deep Q-network based dynamic framing offloading in vehicular edge computing," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 3, pp. 1297–1310, May 2023, doi: [10.1109/TNSE.2022.3172794](https://doi.org/10.1109/TNSE.2022.3172794).
- [45] W. Cheng, X. Liu, X. Wang, and G. Nie, "Task offloading and resource allocation for industrial Internet of Things: A double-dueling deep Q-network approach," *IEEE Access*, vol. 10, pp. 103111–103120, 2022, doi: [10.1109/ACCESS.2022.3210248](https://doi.org/10.1109/ACCESS.2022.3210248).
- [46] H. Van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in *Proc. AAAI Conf. Artif. Intell.*, 2016, vol. 30, no. 1, pp. 1–7.
- [47] C. You, K. Huang, H. Chae, and B.-H. Kim, "Energy-efficient resource allocation for mobile-edge computation offloading," *IEEE Trans. Wireless Commun.*, vol. 16, no. 3, pp. 1397–1411, Mar. 2017.
- [48] L. Yang, M. Li, P. Si, R. Yang, E. Sun, and Y. Zhang, "Energy-efficient resource allocation for blockchain-enabled industrial Internet of Things with deep reinforcement learning," *IEEE Internet Things J.*, vol. 8, no. 4, pp. 2318–2329, Feb. 2021.
- [49] F. Guo, F. R. Yu, H. Zhang, H. Ji, M. Liu, and V. C. M. Leung, "Adaptive resource allocation in future wireless networks with blockchain and mobile edge computing," *IEEE Trans. Wireless Commun.*, vol. 19, no. 3, pp. 1689–1703, Mar. 2020.
- [50] Y. Mao, J. Zhang, S. H. Song, and K. B. Letaief, "Stochastic joint radio and computational resource management for multi-user mobile-edge computing systems," *IEEE Trans. Wireless Commun.*, vol. 16, no. 9, pp. 5994–6009, Sep. 2017.
- [51] Y. Kim, H.-W. Lee, and S. Chong, "Mobile computation offloading for application throughput fairness and energy efficiency," *IEEE Trans. Wireless Commun.*, vol. 18, no. 1, pp. 3–19, Jan. 2019.
- [52] M. Liu, F. R. Yu, Y. Teng, V. C. M. Leung, and M. Song, "Performance optimization for blockchain-enabled industrial Internet of Things (IIoT) systems: A deep reinforcement learning approach," *IEEE Trans. Ind. Informat.*, vol. 15, no. 6, pp. 3559–3570, Jun. 2019.
- [53] R. Tao, W. Liu, X. Chu, and J. Zhang, "An energy saving small cell sleeping mechanism with cell range expansion in heterogeneous networks," *IEEE Trans. Wireless Commun.*, vol. 18, no. 5, pp. 2451–2463, May 2019.



KOMEIL MOGHADDASI (Member, IEEE) was born in West Azerbaijan, Iran, in 1993. He received the bachelor's degree in computer engineering, in 2019. He is currently pursuing the master's degree in information technology with Islamic Azad University, Urmia Branch. His research is centered on reinforcement learning algorithms; cloud, fog, and mobile edge computing; wireless communications; and distributed systems.



SHAKIBA RAJABI (Member, IEEE) was born in West Azerbaijan, Iran, in 1997. She is currently pursuing the master's degree in information technology with the Department of Engineering, Islamic Azad University, Urmia Branch. Her research interests include machine learning algorithms, mobile edge computing, the IoT, and vehicular communications.



FARHAD SOLEIMANIAN GHAREHCHOPOGH was born in Iran, in 1979. He received the B.S. degree in computer engineering from Islamic Azad University, Shabestar Branch, Iran, in 2002, the M.S. degree in computer engineering from Cukurova University, Adana, Turkey, in 2011, and the Ph.D. degree in computer engineering from Hacettepe University, Ankara, in 2015. He has been an Associate Professor of computer engineering with Islamic Azad University, Urmia Branch, Urmia, Iran, since 2015. He is researching metaheuristic algorithms and their application issues. He has published more than 250 papers in international journals and conferences.

...